

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\404-V-001R2.pvdb

Date of Analysis : Aug 11, 2026 3:49pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 4698.000 x 4650.000 mm ID	
Vessel Design Length, Tangent to Tangent	2300.00 cm
Distance of Bottom Tangent above Grade	430.00 cm
Distance of Base above Grade	30.00 cm
Specified Datum Line Distance	0.00 cm
Internal Design Temperature	65 °C
Internal Design Pressure	12.236 kgf/cm ²
External Design Temperature	65 °C
Maximum Allowable Working Pressure	13 kgf/cm ²
Shop Test Pressure	18.399 kgf/cm ²
Field Test Pressure	16.412 kgf/cm ²
Required Minimum Design Metal Temperature	11.4 °C
Warmest Computed Minimum Design Metal Temperature	-8.0 °C
Wind Design Code	IS 875 (Part 3):2015 Amd. 1&2 2020
Earthquake Design Code	Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	+IT
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
headBottom	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom head	12.236	0.00	13.000	3.0000	No	No
Shell-1	12.236	0.00	13.000	3.0000	N/A	No
Shell-2	12.236	0.00	13.000	3.0000	N/A	No
Shell-3	12.236	0.00	13.000	3.0000	N/A	No
Shell-4	12.236	0.00	13.000	3.0000	N/A	No
Shell-5	12.236	0.00	13.000	3.0000	N/A	No
Shell-6	12.236	0.00	13.000	3.0000	N/A	No
Shell-7	12.236	0.00	13.000	3.0000	N/A	No
Shell-8	12.236	0.00	13.000	3.0000	N/A	No
Top head	12.236	0.00	13.000	3.0000	No	No

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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Type	cm	cm	mm	mm	mm	mm		
Skirt	215.0	215.0	12.0	12.0	...	...	0.70	0.70
Skirt	400.0	185.0	12.0	12.0	...	...	0.70	0.80
Ellipse	405.0	5.0	30.0	24.0	23.2	...	1.00	1.00
Cylinder	705.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	1005.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	1305.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	1605.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	1905.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	2205.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	2505.0	300.0	24.0	24.0	23.4	...	1.00	1.00
Cylinder	2695.0	190.0	24.0	24.0	23.4	...	1.00	1.00
Ellipse	2700.0	5.0	30.0	24.0	23.2	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3 100	WRC-107/537	0.867	n/a	Passed
N1 200	WRC-297	0.968	Shell	Passed
N2 150	WRC-107/537	0.867	n/a	Passed

Wind/Earthquake Shear, Bending:

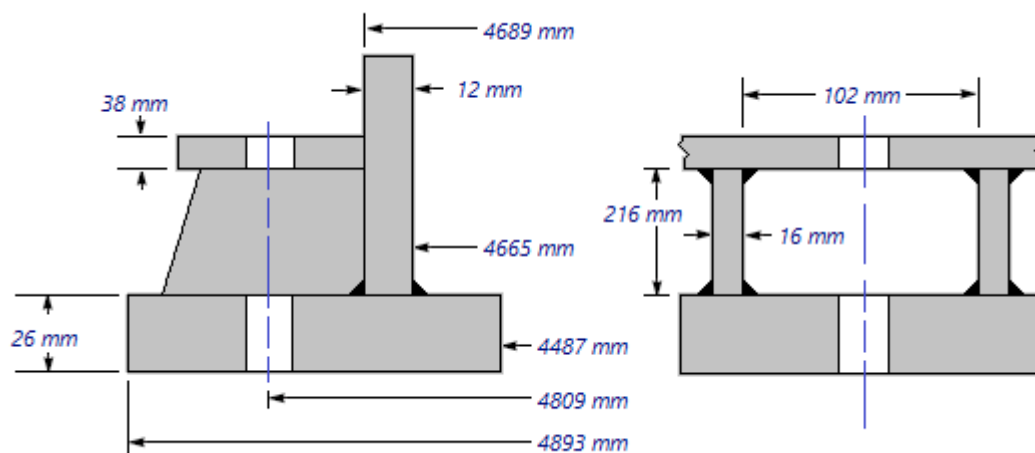
From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	107.5	30391.6	14808.8	472488	302631
20	30	307.5	28786.3	14768.5	408871	271574
30	40	402.5	27405.1	14644.2	356894	245044
40	50	555	27367.7	14314.4	355525	244333
50	60	855	24455.3	13589.4	277790	202425
60	70	1155	21441.3	12791.2	208945	162390
70	80	1455	18222.5	11752.6	149450	124530
80	90	1755	14654.2	10591.9	100134	89559.1
90	100	2055	12125.9	9129.89	59964.2	57927.7
100	110	2355	8528.04	7042.89	28983.1	30591.7
110	120	2600	4771.9	4930.08	9033.21	9467.24
120	130	2697.5	3047.37	2001.53	1604.9	100.077

Abs Max of the all of the Stress Ratio's : 0.6797

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	26.0000	mm
Inside Diameter of Basing	4487.0000	mm
Outside Diameter of Basing	4893.0000	mm
Nominal Diameter of Bolts	42.0000	mm
Diameter of Bolt Circle	4809.0000	mm
Number of Bolts	16	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	159.0000	mm
Height of Gussets	216.0000	mm
Distance between Gussets	102.0000	mm
Thickness of Top Plate or Ring	38.0000	mm
Radial Width of the Top Plate	159.0000	mm
Number of Gussets	32	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	30392.	kgf
Total Earthquake Shear on Support	14809.	kgf
Wind Moment on Support	474288.	kgf-m.
Earthquake Moment on Support	304431.	kgf-m.
Maximum Initial Bolt Load	16729.	kgf

The Wind and Seismic loads were not load factored in this analysis.

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	94319.3	kg
Shop Test - Fabricated + water (full)	514241.9	kg
Shipping - Fab. + rem. intls.+ shipping app.	123638.4	kg
Erected - Fab. + rem. intls.+ insul. (etc)	126833.3	kg
Empty - Fab. + intls. + details + wghts.	126833.3	kg
Operating - Empty + operating liquid (no ca)	126833.3	kg
Field Test - Empty weight + test medium (full)	543560.9	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3 100	13	...	...	OK	8.26	OK	Passed
N1 200	13	...	...	OK	10.16	OK	Passed
M1 600	13	...	...	...		OK	Passed
N2 150	13	...	...	OK	9.22	OK	Passed
N4 100	13	...	...	OK	8.26	OK	Passed
N5 100	13	...	...	OK	8.26	OK	Passed
N6 80	13	...	...	OK	7.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 13 Nozzle : M1 600
 Minimum MAWP Shells/Flanges : 13 Element : Shell-8
 Minimum MAPnc Shells/Flanges : 14 Element : Shell-8

Computed Vessel M.A.W.P. : 13 kgf/cm²

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3 100	0.000	0.000	196.099
40	N1 200	550.000	180.000	399.350
50	M1 600	4574.001	0.000	1231.200
120	N2 150	0.000	0.000	304.659
120	N4 100	0.000	180.000	196.099
120	N5 100	0.000	0.000	196.099
120	N6 80	0.000	0.000	145.299

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N6 80	3.000 in	160	WNF	3.500	11.125	168.90	32.00	542.95	150
N3 100	4.000 in	120	WNF	4.500	11.125	214.30	32.00	274.39	300
N4 100	4.000 in	120	WNF	4.500	11.125	214.30	32.00	558.75	150
N5 100	4.000 in	120	WNF	4.500	11.125	214.30	32.00	558.75	150
N2 150	6.000 in	80	WNF	6.625	10.973	268.27	32.00	524.84	300
N1 200	8.000 in	80	WNF	8.625	12.700	379.08	32.00	276.58	150
M1 600	24.000 in Actual		WNF	24.945	12.000	1273.60	32.00	295.68	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N6 80	SA-106 B	32.000	10.000	24.000	32.000	...
N3 100	SA-106 B	32.000	10.000	24.000	32.000	...
N4 100	SA-106 B	32.000	10.000	24.000	32.000	...
N5 100	SA-106 B	32.000	10.000	24.000	32.000	...
N2 150	SA-106 B	24.000	10.000	24.000	32.000	...
N1 200	SA-106 B	32.000	10.000	24.000	32.000	...
M1 600	SA-516 70	32.000	10.000	24.000	32.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N6 80	...	0.0	500.00	0.00	Top head
N3 100	...	0.0	250.00	0.00	Bottom head
N4 100	...	180.0	500.00	0.00	Top head
N5 100	...	0.0	500.00	0.00	Top head
N2 150	...	0.0	500.00	0.00	Top head
N1 200	455.000	180.0	250.00	0.00	Shell-1
M1 600	857.400	0.0	250.00	0.00	Shell-2

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 30.0mm THK X 4650.0mm ID X 5.0cm	SA-516 70
4	CYLINDER: 24.0mm THK X 4650.0mm ID X 300.0cm	SA-516 70
6	PLATFORM: 1000mm X 1500mm WIDE	
6	LADDER: 4000.0mm	...
1	PLATFORM: 1000mm X 6698mm WIDE	
1	LADDER: 4500.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CLASS 150 GR 1.1, 3.0" BLIND FLANGE(S)	SA-105
1	BASERING: 4893.0mm OD X 4487.0 ID X 26.0mm	SA-516 70
16	BASERING BOLTS: 42.0mm DIA	SA-193 B7
16	BASERING BOLT NUTS: 42.0mm DIA	...
1	BASERING TOP RING: 5007.0mm OD X 4689.0 ID X 38.0mm	...
32	BASERING GUSSET PLATES: 159.0mm X 159.0mm X 216.0mm X 16.0mm	...
6	CYLINDRICAL SKIRT SEGMENT 12.0mm THK X 2154.3mm X 215.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 12.0mm THK X 1729.2mm X 215.0cm	SA-516 70
6	CYLINDRICAL SKIRT SEGMENT 12.0mm THK X 2154.3mm X 185.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 12.0mm THK X 1729.2mm X 185.0cm	SA-516 70
8	CYLINDRICAL SEGMENT 24.0mm THK X 3081.1mm X 300.0cm	SA-516 70
2	CYLINDRICAL SEGMENT 24.0mm THK X 2283.8mm X 300.0cm	SA-516 70
3	CYLINDRICAL SEGMENT 24.0mm THK X 3048.0mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 11560.2mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 6095.9mm X 300.0cm	SA-516 70
3	CYLINDRICAL SEGMENT 24.0mm THK X 3048.0mm X 190.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 11560.2mm X 190.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 6095.9mm X 190.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom head	[10]	B	-2	-2	-29	0.994	24.000	1.00	No
Bottom head	[7]	B	4	-8		0.778	30.000	1.00	No
Shell-1	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-2	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-3	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-4	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-5	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-6	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-7	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Shell-8	[8]	B	-2	-2	-29	1.000	24.000	1.00	No
Top head	[10]	B	-2	-2	-29	0.994	24.000	1.00	No
Top head	[7]	B	4	-8		0.778	30.000	1.00	No
N3 100	[1]	B	-2	-29	-29	0.867	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1 200	[1]	B	-2	-29	-29	0.969	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
M1 600	[1]	B	-2	-29	-29	0.969	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N2 150	[1]	B	-2	-29	-29	0.867	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4 100	[1]	B	-2	-29	-29	0.963	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N5 100	[1]	B	-2	-29	-29	0.963	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N6 80	[1]	B	-2	-29	-29	0.867	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					

Warmest MDMT: 4 -2

Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature -8.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and



UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	12.236	kgf/cm ²
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	User Entered Pressure
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	No
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes



For thickness ratios ≤ 0.35 , MDMT will be -155F (-104C) Yes
For PWHT & P1 Materials the MDMT can be $< -55F (-48C)$ No

Using Metric Material Databases, ASME II D Yes
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/12
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	headBottom
Distance "FROM" to "TO"	215 cm
Skirt Outside Diameter	4689 mm
Diameter of Skirt at Base	4689 mm
Skirt Thickness	12 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	65 °C
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2508.4 kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	10
Detail Type	Weight
Detail ID	Tailing lug
Dist. from "FROM" Node / Offset dist	30 cm
Miscellaneous Weight	1050 kgf
Offset from Element Centerline	2500 mm

Element#	2/12
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	185 cm
Skirt Outside Diameter	4689 mm
Diameter of Skirt at Base	4689 mm
Skirt Thickness	12 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	65 °C
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.8
Weld is pre-Heated	No



Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	5 cm
Inside Diameter	4650 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	12.236 kgf/cm ²
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Nozzle
Detail ID	N3 100
Dist. from "FROM" Node / Offset dist	-48.5 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	22.234 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	4/12
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	300 cm
Element Outside Diameter	4698 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	12.236 kgf/cm ²
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Platform
Detail ID	PL-1
Dist. from "FROM" Node / Offset dist	0 cm
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	55507 cm ²
Platform Weight	5522.9 kgf
Platform Railing Weight	0 kgf/cm
Platform Grating Weight	0.017 kgf/cm ²
Platform Width	1500 mm



Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1 200	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	55.839	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Weight	
Detail ID	N1 Fy wt	
Dist. from "FROM" Node / Offset dist	50	cm
Miscellaneous Weight	2235	kgf
Offset from Element Centerline	3451	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	Cleat weight	
Dist. from "FROM" Node / Offset dist	110	cm
Miscellaneous Weight	2640	kgf
Offset from Element Centerline	0	mm

Element#	5/12	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	300	cm
Inside Diameter	4650	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	12.236	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PL-2	
Dist. from "FROM" Node / Offset dist	100	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	50400	cm ²
Platform Weight	2816.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm



Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M1 600	
Dist. from "FROM" Node / Offset dist	152.4	cm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	572.35	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT

Element From Node	50	
Detail Type	Weight	
Detail ID	Manway davit	
Dist. from "FROM" Node / Offset dist	152.4	cm
Miscellaneous Weight	49.999	kgf
Offset from Element Centerline	0	mm

Element#	6/12	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	300	cm
Inside Diameter	4650	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	12.236	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PL-3	
Dist. from "FROM" Node / Offset dist	200	cm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	50400	cm ²
Platform Weight	2816.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm



Platform Length (top head platform)	0 mm
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Element#	7/12	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	300	cm
Inside Diameter	4650	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	12.236	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516	70
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PL-4	
Dist. from "FROM" Node / Offset dist	300	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	55507	cm ²
Platform Weight	2816.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	300	cm
Inside Diameter	4650	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	12.236	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	300	cm
Inside Diameter	4650	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	12.236	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Platform	
Detail ID	PL-5	
Dist. from "FROM" Node / Offset dist	100	cm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	55507	cm ²
Platform Weight	2816.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	Shell-7	
Distance "FROM" to "TO"	300	cm
Inside Diameter	4650	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	12.236	kgf/cm ²
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Platform	
Detail ID	PL-6	
Dist. from "FROM" Node / Offset dist	200	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	



Platform Wind Area	55507	cm ²
Platform Weight	2816.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element#	11/12
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	Shell-8
Distance "FROM" to "TO"	190 cm
Inside Diameter	4650 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	12.236 kgf/cm ²
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	110
Detail Type	Weight
Detail ID	Trunnion 1
Dist. from "FROM" Node / Offset dist	95 cm
Miscellaneous Weight	999.98 kgf
Offset from Element Centerline	2325 mm

Element From Node	110
Detail Type	Weight
Detail ID	Trunnion 2
Dist. from "FROM" Node / Offset dist	95 cm
Miscellaneous Weight	999.98 kgf
Offset from Element Centerline	2325 mm

Element#	12/12
Element From Node	120
Element To Node	130
Element Type	Elliptical
Description	Top head
Distance "FROM" to "TO"	5 cm
Inside Diameter	4650 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	12.236 kgf/cm ²
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70



Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 120
 Detail Type Platform
 Detail ID PL-7
 Dist. from "FROM" Node / Offset dist 180 cm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 102322 cm²
 Platform Weight 7797 kgf
 Platform Railing Weight 0 kgf/cm
 Platform Grating Weight 0.017 kgf/cm²
 Platform Width 6698 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 3349 mm
 Platform Force Coefficient 1.2
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 cm
 Ladder End Elevation 450 cm
 Unit Weight of Ladder 0.37 kgf/cm
 Platform Length (top head platform) 6698 mm

Element From Node 120
 Detail Type Nozzle
 Detail ID N2 150
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 6 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 52.538 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 120
 Detail Type Nozzle
 Detail ID N4 100
 Dist. from "FROM" Node / Offset dist 2000 mm
 Nozzle Diameter 4 in.
 Nozzle Schedule 120
 Nozzle Class 150
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 29.221 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 120
 Detail Type Nozzle
 Detail ID N5 100
 Dist. from "FROM" Node / Offset dist 2000 mm
 Nozzle Diameter 4 in.
 Nozzle Schedule 120
 Nozzle Class 150
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 29.221 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 120
 Detail Type Nozzle
 Detail ID N6 80
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 3 in.
 Nozzle Schedule 160



Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	25.791	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	120	
Detail Type	Weight	
Detail ID	N2 Fy wt	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	959.98	kgf
Offset from Element Centerline	3425	mm

**XY Coordinate Calculations:**

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
headBottom	...	...	215	...	215
Skirt-1	...	...	400	...	185
Bottom head	...	...	405	...	5
Shell-1	...	...	705	...	300
Shell-2	...	...	1005	...	300
Shell-3	...	...	1305	...	300
Shell-4	...	...	1605	...	300
Shell-5	...	...	1905	...	300
Shell-6	...	...	2205	...	300
Shell-7	...	...	2505	...	300
Shell-8	...	...	2695	...	190
Top head	...	...	2700	...	5

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
headBottom		...	12	1.6	4689	...	1
Skirt-1		...	12	1.6	4689	...	2
Bottom head		12.236	30	3	4650	1407.1	3
Shell-1		12.236	24	3	4698	1407.1	4
Shell-2		12.236	24	3	4650	1407.1	5
Shell-3		12.236	24	3	4650	1407.1	6
Shell-4		12.236	24	3	4650	1407.1	7
Shell-5		12.236	24	3	4650	1407.1	8
Shell-6		12.236	24	3	4650	1407.1	9
Shell-7		12.236	24	3	4650	1407.1	10
Shell-8		12.236	24	3	4650	1407.1	11
Top head		12.236	30	3	4650	1407.1	12

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
headBottom		...	No Calc	No Calc	12	No Calc	1
Skirt-1		...	No Calc	No Calc	12	No Calc	2
Bottom head		12.2362	12.7035	14.5103	24	23.2267	3
Shell-1		12.2362	12.6249	14.4358	24	23.3557	4
Shell-2		12.2362	12.6249	14.4358	24	23.35	5
Shell-3		12.2362	12.6249	14.4358	24	23.35	6
Shell-4		12.2362	12.6249	14.4358	24	23.35	7
Shell-5		12.2362	12.6249	14.4358	24	23.35	8
Shell-6		12.2362	12.6249	14.4358	24	23.35	9
Shell-7		12.2362	12.6249	14.4358	24	23.35	10
Shell-8		12.2362	12.6249	14.4358	24	23.35	11
Top head		12.2362	12.7035	14.5103	24	23.2267	12

Minimum 13 14

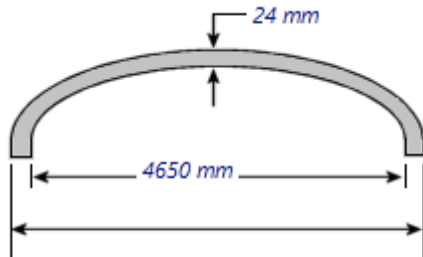
MAWP: 13 kgf/cm², limited by: Shell-8.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		65 °C
Length		5.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress		1355.4 kgf/cm²
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		247870.8 cm²	SG of Contents		0.000
Element Volume		14010405.0 cm³	Weight of Contents		0.0 kgf
Empty Weight		6266.0 kgf	Operating Weight		6266.0 kgf
			Comp. All. Ope. Stress		1084.6 kgf/cm²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 58.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 29.062 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (12.236 \cdot 4656.0 \cdot 0.998) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.236)$$

$$= 20.2267 + 3.0000 = 23.2267 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 21.0) / (0.998 \cdot 4656.0 + 0.2 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 24.0) / (1.0 \cdot 4650.0 + 0.2 \cdot 24.0)$$

$$= 15 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (12.236 \cdot (0.998 \cdot 4656.0 + 0.2 \cdot 21.0)) / (2 \cdot 1.0 \cdot 21.0)$$

$$= 1355.365 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:



$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236) + 3.0 \\ &= 23.350 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 27.0) / (2328.0 + 0.6 \cdot 27.0) \\ &= 16 \text{ kgf/cm}^2 \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (4656.0 / (2 \cdot 1165.5))^2) / 6 \\ &= 0.998285 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.793 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24.0$, $t_r = 20.87$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.994$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 30.0$, $t_r = 21.0$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.778	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-8 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

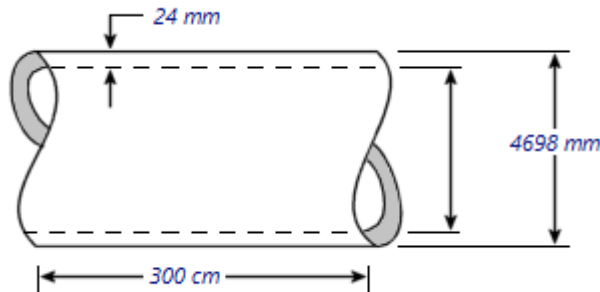
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 58.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 29.062 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Outside Diameter	Do	4698.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (12.236 \cdot 2349.0) / (1407.13 \cdot 1.0 + 0.4 \cdot 12.236)$$

$$= 20.3557 + 3.0000 = 23.3557 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2349.0 - 0.4 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2349.0 - 0.4 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2349.0 - 0.4 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / Ro)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 24.0, t_r = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 1.0, Temp. Reduction = 0 °C

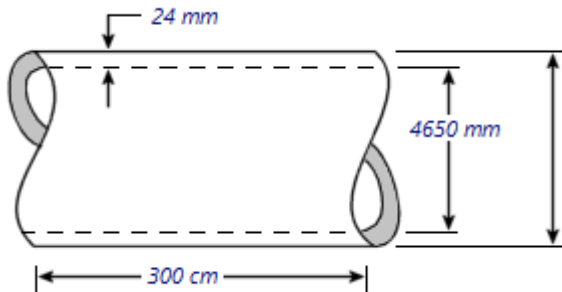


Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C



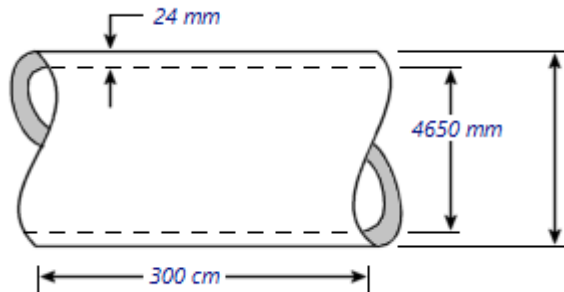
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C



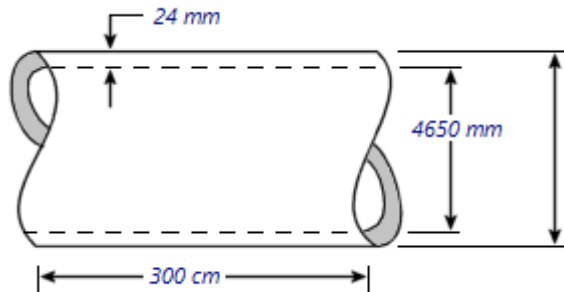
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C



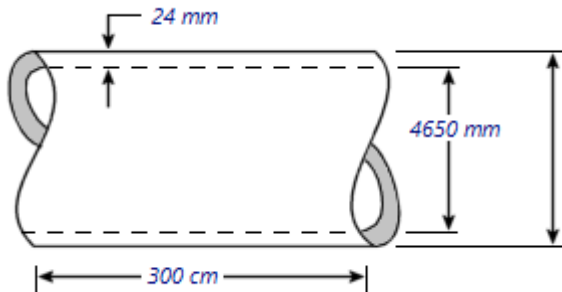
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C



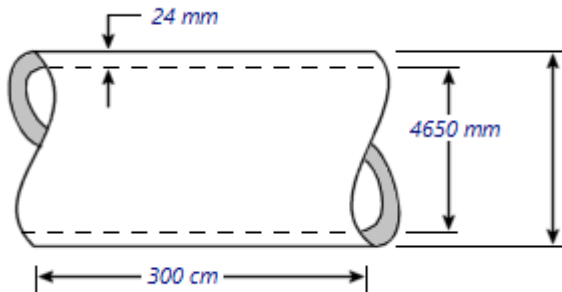
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C



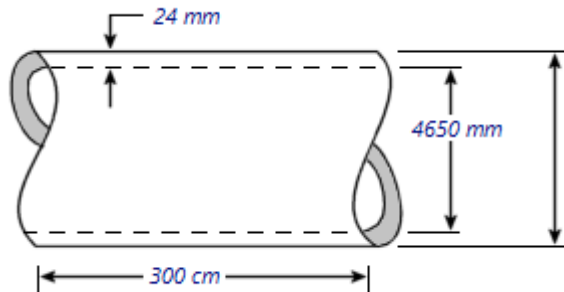
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	4650.000 mm	Actual Stress	1363.8 kgf/cm²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		442776.1 cm²	SG of Contents	0.000	
Element Volume		50946928.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		8193.9 kgf	Operating Weight	8193.9 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C

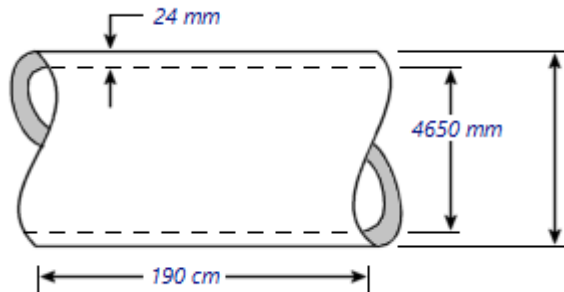


Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	12.236 kgf/cm ²	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		65 °C
Length		190.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	4650.000 mm	Actual Stress		1363.8 kgf/cm ²
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		280424.8 cm ²	SG of Contents		0.000
Element Volume		32266386.0 cm ³	Weight of Contents		0.0 kgf
Empty Weight		5189.5 kgf	Operating Weight		5189.5 kgf
			Comp. All. Ope. Stress		1084.6 kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236)$$

$$= 20.3500 + 3.0000 = 23.3500 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 21.0) / (2328.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 24.0) / (2325.0 + 0.6 \cdot 24.0)$$

$$= 14 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.236 \cdot (2328.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 1363.809 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.513 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 21.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C



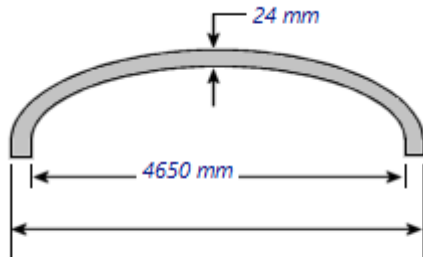
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	12.236 kgf/cm ²	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		5.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	4650.000 mm	Actual Stress	1355.4 kgf/cm ²	
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.	30.000 mm	
Surface Area		247870.8 cm ²	SG of Contents	0.000	
Element Volume		14010405.0 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		6266.0 kgf	Operating Weight	6266.0 kgf	
			Comp. All. Ope. Stress	1084.6 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 58.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 29.062 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (12.236 \cdot 4656.0 \cdot 0.998) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.236)$$

$$= 20.2267 + 3.0000 = 23.2267 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 21.0) / (0.998 \cdot 4656.0 + 0.2 \cdot 21.0)$$

$$= 13 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 24.0) / (1.0 \cdot 4650.0 + 0.2 \cdot 24.0)$$

$$= 15 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (12.236 \cdot (0.998 \cdot 4656.0 + 0.2 \cdot 21.0)) / (2 \cdot 1.0 \cdot 21.0)$$

$$= 1355.365 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:



$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (12.236 \cdot 2328.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.236) + 3.0 \\ &= 23.350 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 27.0) / (2328.0 + 0.6 \cdot 27.0) \\ &= 16 \text{ kgf/cm}^2 \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (4656.0 / (2 \cdot 1165.5))^2) / 6 \\ &= 0.998285 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.793 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24.0$, $t_r = 20.87$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.994$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 30.0$, $t_r = 21.0$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.625 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.778	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-8 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 58.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 29.062 mm

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	16.412 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	15.907 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	18.302 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	13.887 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	17.498 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	17.498 kgf/cm ²
App 27-4	= M.A.W.P.	12.625 kgf/cm ²

User Defined Hydrostatic Test Pressure at High Point 18.399 kgf/cm²

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	1829.3	2386.0	0.767	18.86
Shell-1	1839.2	2386.0	0.771	18.87
Shell-2	1838.8	2386.0	0.771	18.86
Shell-3	1838.8	2386.0	0.771	18.86
Shell-4	1838.8	2386.0	0.771	18.86
Shell-5	1838.8	2386.0	0.771	18.86
Shell-6	1838.8	2386.0	0.771	18.86
Shell-7	1838.8	2386.0	0.771	18.86
Shell-8	1838.8	2386.0	0.771	18.86
Top head	1829.3	2386.0	0.767	18.86

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3 100	Nozzle	1202.20	1202.20	1.000
N3 100	Pad	1406.14	1406.14	1.000
N1 200	Nozzle	1202.20	1202.20	1.000
N1 200	Pad	1406.14	1406.14	1.000
M1 600	Nozzle	1406.14	1406.14	1.000
M1 600	Pad	1406.14	1406.14	1.000
N2 150	Nozzle	1202.20	1202.20	1.000
N2 150	Pad	1406.14	1406.14	1.000
N4 100	Nozzle	1202.20	1202.20	1.000
N4 100	Pad	1406.14	1406.14	1.000
N5 100	Nozzle	1202.20	1202.20	1.000
N5 100	Pad	1406.14	1406.14	1.000
N6 80	Nozzle	1202.20	1202.20	1.000
N6 80	Pad	1406.14	1406.14	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	1407.13	1407.13	1.000
Shell-1	1407.13	1407.13	1.000
Shell-2	1407.13	1407.13	1.000
Shell-3	1407.13	1407.13	1.000
Shell-4	1407.13	1407.13	1.000
Shell-5	1407.13	1407.13	1.000
Shell-6	1407.13	1407.13	1.000
Shell-7	1407.13	1407.13	1.000
Shell-8	1407.13	1407.13	1.000
Top head	1407.13	1407.13	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N3 100	89.36	2214.75	0.040
N1 200	155.20	2214.75	0.070
M1 600	490.47	2404.41	0.204
N2 150	137.10	2214.75	0.062
N4 100	89.36	2214.75	0.040
N5 100	89.36	2214.75	0.040
N6 80	67.82	2214.75	0.031

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	16.412 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	15.907 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	18.455 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	13.887 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	17.498 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	17.498 kgf/cm ²
App 27-4	= M.A.W.P.	12.625 kgf/cm ²

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 12.625 • 1.0

= 16.412 kgf/cm²

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	2098.3	2386.0	0.879	18.94
Shell-1	2097.9	2386.0	0.879	18.82
Shell-2	2064.5	2386.0	0.865	18.52
Shell-3	2031.1	2386.0	0.851	18.22
Shell-4	1997.6	2386.0	0.837	17.92
Shell-5	1964.2	2386.0	0.823	17.62
Shell-6	1930.8	2386.0	0.809	17.32
Shell-7	1897.4	2386.0	0.795	17.02
Shell-8	1863.9	2386.0	0.781	16.72
Top head	1831.4	2386.0	0.768	16.53

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3 100	Nozzle	1202.20	1202.20	1.000
N3 100	Pad	1406.14	1406.14	1.000
N1 200	Nozzle	1202.20	1202.20	1.000
N1 200	Pad	1406.14	1406.14	1.000
M1 600	Nozzle	1406.14	1406.14	1.000
M1 600	Pad	1406.14	1406.14	1.000
N2 150	Nozzle	1202.20	1202.20	1.000
N2 150	Pad	1406.14	1406.14	1.000
N4 100	Nozzle	1202.20	1202.20	1.000
N4 100	Pad	1406.14	1406.14	1.000
N5 100	Nozzle	1202.20	1202.20	1.000
N5 100	Pad	1406.14	1406.14	1.000
N6 80	Nozzle	1202.20	1202.20	1.000
N6 80	Pad	1406.14	1406.14	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	1407.13	1407.13	1.000
Shell-1	1407.13	1407.13	1.000
Shell-2	1407.13	1407.13	1.000
Shell-3	1407.13	1407.13	1.000
Shell-4	1407.13	1407.13	1.000
Shell-5	1407.13	1407.13	1.000
Shell-6	1407.13	1407.13	1.000



Shell-7	1407.13	1407.13	1.000
Shell-8	1407.13	1407.13	1.000
Top head	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N3 100	89.74	2214.75	0.041
N1 200	154.82	2214.75	0.070
M1 600	481.59	2404.41	0.200
N2 150	120.16	2214.75	0.054
N4 100	78.32	2214.75	0.035
N5 100	78.32	2214.75	0.035
N6 80	59.45	2214.75	0.027

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	400	...	...	No Calc	No Calc
20	30	400	...	...	No Calc	No Calc
30	40	No Calc	4698	21	0.00062083	632.908
40	50	2377.5	4698	21	0.73223E-04	74.6474
50	60	2377.5	4698	21	0.73223E-04	74.6474
60	70	2377.5	4698	21	0.73223E-04	74.6474
70	80	2377.5	4698	21	0.73223E-04	74.6474
80	90	2377.5	4698	21	0.73223E-04	74.6474
90	100	2377.5	4698	21	0.73223E-04	74.6474
100	110	2377.5	4698	21	0.73223E-04	74.6474
110	120	2377.5	4698	21	0.73223E-04	74.6474
120	130	No Calc	4698	21	0.00062083	632.908

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	Design External Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	24	No Calc	...	3.14343
40	50	24	No Calc	...	0.4449
50	60	24	No Calc	...	0.4449
60	70	24	No Calc	...	0.4449
70	80	24	No Calc	...	0.4449
80	90	24	No Calc	...	0.4449
90	100	24	No Calc	...	0.4449
100	110	24	No Calc	...	0.4449
110	120	24	No Calc	...	0.4449
120	130	24	No Calc	...	3.14343

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	400	No Calc	No Calc	No Calc
20	30	400	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	2377.5	No Calc	No Calc	No Calc
50	60	2377.5	No Calc	No Calc	No Calc
60	70	2377.5	No Calc	No Calc	No Calc
70	80	2377.5	No Calc	No Calc	No Calc
80	90	2377.5	No Calc	No Calc	No Calc
90	100	2377.5	No Calc	No Calc	No Calc
100	110	2377.5	No Calc	No Calc	No Calc
110	120	2377.5	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	4390.27	...	3998.55	...	307.319
20	30	2528.08	...	2191.04	...	176.966
30	40	6265.96	14010405	5639.36	14063611	438.617
40	50	8193.94	50946928	7174.27	51078492	573.576
50	60	8193.94	50946928	7174.27	51078492	573.576
60	70	8193.94	50946928	7174.27	51078492	573.576
70	80	8193.94	50946928	7174.27	51078492	573.576
80	90	8193.94	50946928	7174.27	51078492	573.576
90	100	8193.94	50946928	7174.27	51078492	573.576
100	110	8193.94	50946928	7174.27	51078492	573.576
110	120	5189.5	32266386	4543.71	32349712	363.265
120	130	6265.96	14010405	5639.36	14063611	438.617
Total		81997	0.417E+09	72231	0.418E+09	5739

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
10	Wght	1049.98	250	30	...	Tailing lug
30	Nozl	23.7902	-4.85	-132.551	...	N3 100
40	Plat	5909.49	0.29767E-05	...	...	PL-1
40	Nozl	59.7481	-245.001	50	...	N1 200
40	Wght	2234.96	-345.1	50	...	N1 Fy wt
40	Wght	2639.95	...	110	...	Cleat weight
50	Plat	3013.34	...	100	-214.014	PL-2
50	Nozl	612.414	245.001	152.4	...	M1 600
50	Wght	49.9991	...	152.4	...	Manway davit
60	Plat	3013.34	...	200	214.014	PL-3
70	Plat	3013.34	...	300	-214.014	PL-4
90	Plat	3013.34	...	100	214.014	PL-5
100	Plat	3013.34	...	200	-214.014	PL-6
110	Wght	999.982	232.5	95.0002	...	Trunnion 1
110	Wght	999.982	-232.5	95.0002	...	Trunnion 2
120	Plat	8342.82	...	180	...	PL-7
120	Nozl	56.2157	...	145.05	...	N2 150
120	Nozl	31.2662	-200	88.0795	...	N4 100
120	Nozl	31.2662	200	88.0795	...	N5 100
120	Nozl	27.596	150	117.62	...	N6 80
120	Wght	959.983	342.5	...	...	N2 Fy wt

Total Weight of Each Detail Type:

Platforms	29319.0
Nozzles	842.3
Weights	8934.8

Sum of the Detail Weights 39096.2 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	87737.1	87737.1	87737.1	87737.1	87737.1	87737.1
Nozzles	842.3	842.3	842.3	842.3	842.3	842.3
Wld Weights	5739.9	5739.9	5739.9	5739.9	5739.9	5739.9
Platforms	...	...	29319.0	29319.0	29319.0	29319.0
Empty Weights	...	...	...	3194.9	3194.9	...
Test Weights	...	3194.9	...	...	...	...
Ope Weights	...	...	...	...	...	3194.9



Test Liquid	...	416727.7	...	...	...	...
Totals	94319.3	514241.9	123638.4	126833.3	126833.3	126833.3

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	94319.3 kg
Shop Test Wt.	- Fabricated weight + water (full)	514241.9 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	123638.4 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	126833.3 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	126833.3 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	126833.3 kg
Field Test Wt.	- Empty weight + test medium (full)	543560.9 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	46411.4 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	213236.3 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	50336.8 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	316715	315094
20	30	272522	271127
30	40	247871	241689
40	50	442776	438252
50	60	442776	438252
60	70	442776	438252
70	80	442776	438252
80	90	442776	438252
90	100	442776	438252
100	110	442776	438252
110	120	280425	277560
120	130	247871	241689

Total	4464836.5	4414924.5 cm ²
Total	4805.9	4752.2 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	5747.57	5747.57	5747.57	...	5747.57
20	30	2705.05	2705.05	2705.05	...	2705.05
30	40	6728.36	6728.36	20732.4	...	6728.36
40	50	19611.7	19611.7	70535.6	...	19611.7
50	60	12443.3	12443.3	63367.2	6449	12443.3
60	70	11780.9	11780.9	62704.8	6449	11780.9
70	80	11780.9	11780.9	62704.8	6449	11780.9
80	90	8767.51	8767.51	59691.5	...	8767.51
90	100	11780.9	11780.9	62704.8	6449	11780.9
100	110	11780.9	11780.9	62704.8	6449	11780.9
110	120	7552.73	7552.73	39804.6	...	7552.73
120	130	16153.7	16153.7	30157.8	...	16153.7



Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	126833	126833	543561
20	30	121086	121086	537813
30	40	118381	118381	535108
40	50	111652	111652	514376
50	60	92040.7	92040.7	443840
60	70	79597.4	79597.4	380473
70	80	67816.5	67816.5	317768
80	90	56035.7	56035.7	255063
90	100	47268.2	47268.2	195372
100	110	35487.3	35487.3	132667
110	120	23706.5	23706.5	69962.4
120	130	16153.7	16153.7	30157.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	32245	32245	32245
20	30	32245	32245	32245
30	40	32245	32245	32245
40	50	32245	32245	32245
50	60	32245	32245	32245
60	70	25796	25796	25796
70	80	19347	19347	19347
80	90	12898	12898	12898
90	100	12898	12898	12898
100	110	6449	6449	6449
110	120	...	...	...
120	130	...	...	...



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3 100	49.8	52.0	65	300	GR 1.1	41.2	33.6	49.8	33.6
N1 200	19.0	20.0	65	150	GR 1.1	28.2	13.7	19.0	0.4
M1 600	19.05	19.99	65	150	GR 1.1	...	...	...	...
N2 150	49.8	52.0	65	300	GR 1.1	32.3	42.4	49.8	42.4
N4 100	19.05	19.99	65	150	GR 1.1	...	...	...	...
N5 100	19.05	19.99	65	150	GR 1.1	...	...	...	...
N6 80	19.05	19.99	65	150	GR 1.1	...	...	...	...
Min Rating	13.721	19.987, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N3 100	138.176	0.50	204	275	12.6	33.6	Pass
N1 200	248.539	1.20	815	551	12.6	13.7	Pass
N2 150	196.088	0.50	459	408	12.6	42.4	Pass

Evaluating Flange on Nozzle [N3 100]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(204) + 4 \cdot \text{abs}(275)138.176 \leq \pi \cdot 138.176^3 \cdot [(49.835 - 12.625) + .500 \cdot 49.835]$$

$$= 0.34141E+04 \leq 0.51492E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3 100 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(49.835, 0.500 \cdot 49.835 + 49.835 - 3414.1 / (\pi \cdot 138.176^3))$$

$$= \min(49.835, 33.560)$$

$$= 33.560 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N1 200]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(815) + 4 \cdot \text{abs}(551)248.539 \leq \pi \cdot 248.539^3 \cdot [(19.049 - 12.625) + 1.200 \cdot 19.049]$$

$$= 0.13595E+05 \leq 0.14124E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 200 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(19.049, 1.200 \cdot 19.049 + 19.049 - 13595.2 / (\pi \cdot 248.539^3))$$

$$= \min(19.049, 13.721)$$

$$= 13.721 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N2 150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(459) + 4 \cdot \text{abs}(408)196.088 \leq \pi \cdot 196.088^3 \cdot [(49.835 - 12.625) + .500 \cdot 49.835]$$

$$= 0.76593E+04 \leq 0.14716E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2 150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(49.835, 0.500 \cdot 49.835 + 49.835 - 7659.3 / (\pi \cdot 196.088^3))$$

$$= \min(49.835, 42.417)$$

$$= 42.417 \text{ kgf/cm}^2$$

PV Elite 28 U2 Licensee: SL User
FileName : 404-V-001_R2
Nozzle Flange MAWP:

Step: 18 3:50pm Aug 11,2026

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HEXAGON

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 3.2055 Hz.

The Natural Frequency for the Vessel (Ope...) is 3.2055 Hz.

The Natural Frequency for the Vessel (Filled) is 1.5821 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 6.001

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 3.206 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (6.001) is > 5 .

Note:

As per Section 9.1, Dynamic effects of wind should be considered. The Cross-Wind dynamic effects can be considered using the Vortex Shedding which can be invoked from Design Constraints tab. While the Along-Wind effects can be considered using Gust Response Factor method, which can be invoked from the Wind Load tab.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_{2,1} \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1983.503, 1.0 \cdot 0.8 \cdot 1.0 \cdot 0.6 \cdot 57.5^2)$$

$$= 1586.8 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 12.1 \cdot 0.82 \cdot 1586.803$$

$$= 15741.7 \text{ N}$$

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	1.38	1.0000	1.0000	1.0000	57.50	0.8200
20	3.38	1.0000	1.0000	1.0000	57.50	0.8200
30	4.32	1.0000	1.0000	1.0000	57.50	0.8200
40	5.85	1.0000	1.0000	1.0000	57.50	0.8200
50	8.85	1.0000	1.0000	1.0000	57.50	0.8200
60	11.85	1.0000	1.0185	1.0000	58.56	0.8200
70	14.85	1.0000	1.0485	1.0000	60.28	0.8200
80	17.85	1.0000	1.0614	1.0000	61.03	0.8200
90	20.85	1.0000	1.0743	1.0000	61.77	0.8200
100	23.85	1.0000	1.0893	1.0000	62.63	0.8200
110	26.30	1.0000	1.1015	1.0000	63.33	0.8200



120 27.78 1.0000 1.1089 1.0000 63.76 0.8200

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, cm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, cm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure cm
 Lc - Total length of conical section(s) of vessel, cm
 tb - Un corroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.00032 (High); 0.00032 < 0.0004 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0.0/2700.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[15.41/88.58^2] \cdot 10^4 = 19.637$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.0004 no chance. [Vp]:

= $W / (L \cdot Dr^2)$
 = $116384 / (2700.0 \cdot 469.8^2)$
 = 0.19530E-03

Compute the damping factor Df which is a measure of instability [Df]:

= $W \cdot D1 / (L \cdot Dr^2)$
 = $116384 \cdot 0.03 / (2700.0 \cdot 469.8^2)$
 = 0.58590E-05

Compute the critical wind velocity [Vc]:

= $3.4 \cdot f \cdot Dr$
 = $3.4 \cdot 3.206 \cdot 15.413$
 = 167.986 mile/hr [270.340 km/hr]

Compute the velocity at the top of the tower [Vw]:

= $V30 \cdot (L / (30 + BaseHeight))^{0.143}$
 = $111.85 \cdot (88.58 / (30 + 1.0))^{0.143}$
 = 129.979 mile/hr [209.175 km/hr]

Compute the maximum gust velocity using the gust response factor Gh [Vg]:

= $Vw \cdot Gh$
 = $209.175 \cdot 1.0$
 = 209.175 km/hr

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf	Cf
PL-1	55507.20	435.00	161.88	898.52	1.20



PL-2	50400.00	835.00	161.88	815.85	1.20
PL-3	50400.00	1235.00	169.59	854.75	1.20
PL-4	55507.20	1635.00	180.16	1000.02	1.20
PL-5	55507.20	2035.00	186.07	1032.80	1.20
PL-6	55507.20	2435.00	192.95	1070.99	1.20
PL-7	102322.00	2905.00	200.37	2050.21	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Tailing lug	0.00	60.00	161.88	0.00
N1 Fy wt	0.00	485.00	161.88	0.00
Cleat weight	0.00	545.00	161.88	0.00
Manway davit	0.00	887.40	161.92	0.00
Trunnion 1	0.00	2630.00	196.40	0.00
Trunnion 2	0.00	2630.00	196.40	0.00
N2 Fy wt	0.00	2725.00	198.10	0.00

The Natural Frequency for the Vessel (Ope...) is 3.2055 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	137.5	562.68	120976	161.875	1605.28
20	30	337.5	562.68	104096	161.875	1381.29
30	40	432.5	563.76	2818.8	161.875	37.4038
40	50	585	563.76	169128	161.875	2912.36
50	60	885	563.76	169128	161.875	3013.99
60	70	1185	563.76	169128	167.92	3218.85
70	80	1485	563.76	169128	177.958	3568.23
80	90	1785	563.76	169128	182.364	2528.28
90	100	2085	563.76	169128	186.806	3597.9
100	110	2385	563.76	169128	192.059	3756.14
110	120	2630	563.76	107114	196.404	1724.52
120	130	2777.67	563.76	55354.2	199.045	3047.37



Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Computed Natural Frequencies (OPE): 7

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	3.2770	20.5898	0.3052
2	18.8129	118.2049	0.0532
3	28.8335	181.1664	0.0347
4	42.4859	266.9470	0.0235
5	71.5319	449.4485	0.0140
6	82.7593	519.9920	0.0121
7	101.8541	639.9679	0.0098

Mass Participation Factors: 7

Mode	X	Y
1	3.44499193	0.00000000
2	0.04447348	-0.00000000
3	-0.00000000	0.01954914
4	0.00311611	0.00000000
5	0.00041330	0.00000000
6	0.00000000	0.00054020
7	0.00009841	-0.00000000

Computed EigenVectors, Mass Normalized: 7

Node	Dx	Dy	Rz
Mode # 1	Freq (HZ) = 3.2770		
10	0.000000	0.000000	-0.000000
20	0.001391	0.000000	-0.000020
30	0.003843	0.000000	-0.000035
40	0.003918	0.000000	-0.000035
50	0.009124	0.000000	-0.000046



60	0.015523	0.000000	-0.000055
70	0.022871	0.000000	-0.000063
80	0.030931	0.000000	-0.000068
90	0.039488	0.000000	-0.000072
100	0.048345	0.000000	-0.000074
110	0.057325	0.000000	-0.000075
120	0.063013	0.000000	-0.000075
130	0.063161	0.000000	-0.000075

Mode # 2 Freq (HZ) = 18.8129

10	0.000000	-0.000000	-0.000001
20	0.013871	-0.000000	-0.000060
30	0.028361	-0.000000	-0.000078
40	0.028640	-0.000000	-0.000078
50	0.043501	-0.000000	-0.000063
60	0.052371	-0.000000	-0.000020
70	0.053041	-0.000000	0.000037
80	0.044551	-0.000000	0.000096
90	0.027831	-0.000000	0.000147
100	0.004848	-0.000000	0.000184
110	-0.021615	-0.000000	0.000203
120	-0.038642	-0.000000	0.000206
130	-0.039065	-0.000000	0.000206

Mode # 3 Freq (HZ) = 28.8335

10	-0.000000	0.000001	0.000000
20	-0.000000	0.009390	-0.000000
30	-0.000000	0.017408	-0.000000
40	-0.000000	0.017522	-0.000000
50	0.000000	0.023959	-0.000000
60	-0.000000	0.029784	0.000000
70	-0.000000	0.035034	-0.000000
80	0.000000	0.039627	-0.000000
90	0.000000	0.043570	0.000000
100	-0.000000	0.046800	0.000000
110	-0.000000	0.049151	-0.000000
120	0.000000	0.050160	-0.000000
130	0.000000	0.050171	-0.000000

Mode # 4 Freq (HZ) = 42.4859

10	0.000000	0.000000	-0.000001
20	0.030307	0.000000	-0.000032
30	0.053082	0.000000	0.000024
40	0.053276	0.000000	0.000025
50	0.047282	0.000000	0.000112
60	0.018243	0.000000	0.000174
70	-0.017918	0.000000	0.000163
80	-0.044439	0.000000	0.000078
90	-0.049941	-0.000000	-0.000048
100	-0.031270	-0.000000	-0.000168
110	0.005667	-0.000000	-0.000240
120	0.031331	-0.000000	-0.000252
130	0.031904	-0.000000	-0.000252

Mode # 5 Freq (HZ) = 71.5319

10	0.000001	0.000000	-0.000000
20	0.034469	0.000000	0.000024
30	0.051625	0.000000	0.000139
40	0.051470	0.000000	0.000141
50	0.002648	0.000000	0.000201
60	-0.046541	0.000000	0.000109
70	-0.049974	-0.000000	-0.000064
80	-0.006063	-0.000000	-0.000160
90	0.042777	-0.000000	-0.000091
100	0.053236	-0.000000	0.000092
110	0.010433	-0.000000	0.000244
120	-0.026326	0.000000	0.000274
130	-0.027049	0.000000	0.000274

Mode # 6 Freq (HZ) = 82.7593



10	0.000000	0.000002	-0.000000
20	0.000001	0.024064	0.000000
30	-0.000000	0.043491	0.000000
40	-0.000000	0.043726	0.000000
50	0.000000	0.050274	0.000000
60	0.000000	0.046240	0.000000
70	-0.000000	0.034853	-0.000000
80	0.000000	0.018073	0.000000
90	-0.000000	-0.001145	-0.000000
100	0.000000	-0.020208	-0.000000
110	-0.000000	-0.036145	-0.000000
120	0.000000	-0.043333	-0.000000
130	0.000000	-0.043409	-0.000000

Mode # 7 Freq (HZ) = 101.8541

10	0.000001	-0.000000	-0.000000
20	0.035131	-0.000000	0.000045
30	0.044006	-0.000000	0.000171
40	0.043548	-0.000000	0.000173
50	-0.043760	-0.000000	0.000127
60	-0.040102	-0.000000	-0.000082
70	0.030894	0.000000	-0.000123
80	0.055545	0.000000	0.000036
90	0.003069	0.000000	0.000137
100	-0.051420	0.000000	0.000012
110	-0.021731	-0.000000	-0.000178
120	0.019462	-0.000000	-0.000223
130	0.020180	-0.000000	-0.000223

Mass Percentages:

Included(X)	Included(Y)	Added(X)	Added(Y)
97.70	96.01	2.30	3.99

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:

Node	Fx(kgf)	Fy(kgf)	Mz(kgf-m.)
10	14810.1	7307.9	302630.9
Max	14307.9	7285.8	302236.7
Contrib	1 (X)	3 (Y)	1 (X)

Element Forces and Moments:

Node	Fx(kgf)	Fy(kgf)	Mz(kgf-m.)
10	14808.8	7305.3	302630.9
Max	14307.9	7285.8	302236.7
Contrib	1 (X)	3 (Y)	1 (X)
20	14808.8	7305.3	271573.5
Max	14307.9	7285.8	271474.6
Contrib	1 (X)	3 (Y)	1 (X)
20	14768.5	7247.5	271573.5
Max	14289.5	7231.2	271474.6
Contrib	1 (X)	3 (Y)	1 (X)
30	14768.5	7247.5	245043.9
Max	14289.5	7231.2	245038.9
Contrib	1 (X)	3 (Y)	1 (X)
30	14644.2	7106.4	245043.9
Max	14221.0	7094.8	245038.9
Contrib	1 (X)	3 (Y)	1 (X)
40	14644.2	7106.4	244332.7
Max	14221.0	7094.8	244327.9
Contrib	1 (X)	3 (Y)	1 (X)



40	14314.4	6715.5	244332.7
Max	14025.8	6711.3	244327.9
Contrib	1 (X)	3 (Y)	1 (X)
50	14314.4	6715.5	202424.6
Max	14025.8	6711.3	202250.5
Contrib	1 (X)	3 (Y)	1 (X)
50	13589.4	6076.2	202424.6
Max	13472.6	6073.1	202250.5
Contrib	1 (X)	3 (Y)	1 (X)
60	13589.4	6076.2	162390.2
Max	13472.6	6073.1	161832.6
Contrib	1 (X)	3 (Y)	1 (X)
60	12791.2	5484.3	162390.2
Max	12761.4	5473.6	161832.6
Contrib	1 (X)	3 (Y)	1 (X)
70	12791.2	5484.3	124529.6
Max	12761.4	5473.6	123548.3
Contrib	1 (X)	3 (Y)	1 (X)
70	11752.6	4812.1	124529.6
Max	11742.2	4787.7	123548.3
Contrib	1 (X)	3 (Y)	1 (X)
80	11752.6	4812.1	89559.1
Max	11742.2	4787.7	88321.6
Contrib	1 (X)	3 (Y)	1 (X)
80	10591.9	4148.6	89559.1
Max	10540.2	4111.2	88321.6
Contrib	1 (X)	3 (Y)	1 (X)
90	10591.9	4148.6	57927.7
Max	10540.2	4111.2	56701.0
Contrib	1 (X)	3 (Y)	1 (X)
90	9129.9	3413.0	57927.7
Max	9005.5	3367.3	56701.0
Contrib	1 (X)	3 (Y)	1 (X)
100	9129.9	3413.0	30591.7
Max	9005.5	3367.3	29684.4
Contrib	1 (X)	3 (Y)	1 (X)
100	7042.9	2495.7	30591.7
Max	6851.1	2451.0	29684.4
Contrib	1 (X)	3 (Y)	1 (X)
110	7042.9	2495.7	9467.2
Max	6851.1	2451.0	9131.0
Contrib	1 (X)	3 (Y)	1 (X)
110	4930.1	1695.0	9467.2
Max	4755.0	1661.4	9131.0
Contrib	1 (X)	3 (Y)	1 (X)
120	4930.1	1695.0	100.1
Max	4755.0	1661.4	96.5
Contrib	1 (X)	3 (Y)	1 (X)
120	2001.5	687.0	100.1
Max	1929.7	673.4	96.5
Contrib	1 (X)	3 (Y)	1 (X)
130	2001.5	687.0	0.0
Max	1929.7	673.4	0.0
Contrib	1 (X)	3 (Y)	

Displacements:

Node	Dx(mm)	Dy(mm)	Rz(Deg)
10	0.0000	0.0000	0.0000
Max	0.0000	0.0000	0.0000
Contrib	1 (X)	3 (Y)	1 (X)



20	0.1228	0.0047	0.0039
Max	0.1217	0.0047	0.0039
Contrib	1 (X)	3 (Y)	1 (X)
30	0.3378	0.0087	0.0069
Max	0.3363	0.0086	0.0068
Contrib	1 (X)	3 (Y)	1 (X)
40	0.3444	0.0087	0.0069
Max	0.3429	0.0087	0.0069
Contrib	1 (X)	3 (Y)	1 (X)
50	0.7999	0.0119	0.0091
Max	0.7984	0.0119	0.0091
Contrib	1 (X)	3 (Y)	1 (X)
60	1.3596	0.0148	0.0109
Max	1.3583	0.0148	0.0109
Contrib	1 (X)	3 (Y)	1 (X)
70	2.0021	0.0174	0.0124
Max	2.0013	0.0174	0.0124
Contrib	1 (X)	3 (Y)	1 (X)
80	2.7070	0.0197	0.0134
Max	2.7065	0.0197	0.0134
Contrib	1 (X)	3 (Y)	1 (X)
90	3.4554	0.0216	0.0141
Max	3.4553	0.0216	0.0141
Contrib	1 (X)	3 (Y)	1 (X)
100	4.2303	0.0232	0.0146
Max	4.2303	0.0232	0.0146
Contrib	1 (X)	3 (Y)	1 (X)
110	5.0162	0.0244	0.0148
Max	5.0161	0.0244	0.0148
Contrib	1 (X)	3 (Y)	1 (X)
120	5.5140	0.0249	0.0148
Max	5.5138	0.0249	0.0148
Contrib	1 (X)	3 (Y)	1 (X)
130	5.5270	0.0249	0.0148
Max	5.5268	0.0249	0.0148
Contrib	1 (X)	3 (Y)	1 (X)



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	107.5	30391.6	14808.8	472488	302631
20	30	307.5	28786.3	14768.5	408871	271574
30	40	402.5	27405.1	14644.2	356894	245044
40	50	555	27367.7	14314.4	355525	244333
50	60	855	24455.3	13589.4	277790	202425
60	70	1155	21441.3	12791.2	208945	162390
70	80	1455	18222.5	11752.6	149450	124530
80	90	1755	14654.2	10591.9	100134	89559.1
90	100	2055	12125.9	9129.89	59964.2	57927.7
100	110	2355	8528.04	7042.89	28983.1	30591.7
110	120	2600	4771.9	4930.08	9033.21	9467.24
120	130	2697.5	3047.37	2001.53	1604.9	100.077

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.000; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	30391.6	0.02904	0.11348	0.00010308
20	30	28786.3	0.22745	0.37707	0.00018015
30	40	27405.1	0.38158	0.38611	0.00018118
40	50	27367.7	0.68022	1.01558	0.00023621
50	60	24455.3	1.38725	1.79059	0.00027847
60	70	21441.3	2.22126	2.6752	0.00030954
70	80	18222.5	3.1487	3.63835	0.00033112
80	90	14654.2	4.1411	4.65422	0.00034496
90	100	12125.9	5.17518	5.70182	0.00035255
100	110	8528.04	6.23227	6.76505	0.00035569
110	120	4771.9	7.10315	7.44152	0.00035625
120	130	3047.37	7.45043	7.45933	0.00035625

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	323.787	2023.67
20	30	323.787	2023.67
30	40	324.408	2027.55
40	50	324.408	2027.55
50	60	324.408	2027.55
60	70	324.408	2027.55
70	80	324.408	2027.55
80	90	324.408	2027.55
90	100	324.408	2027.55
100	110	324.408	2027.55
110	120	324.408	2027.55
120	130	324.408	2027.55

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 135.000 Actual deflection : 7.459 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	1763.18	1528.12	205635536	178279936
20	30	1763.18	1528.12	205635536	178279936
30	40	3524.11	3085.57	409701664	359176704
40	50	3524.11	3085.57	409701664	359176704
50	60	3524.11	3085.57	409701664	359176704
60	70	3524.11	3085.57	409701664	359176704
70	80	3524.11	3085.57	409701664	359176704
80	90	3524.11	3085.57	409701664	359176704
90	100	3524.11	3085.57	409701664	359176704
100	110	3524.11	3085.57	409701664	359176704
110	120	3524.11	3085.57	409701664	359176704
120	130	3524.11	3085.57	409701664	359176704



Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1181.99	2004.25	-678.562	-678.562
20	30	1350.85	2290.57	-678.562	-678.562
30	40	1688.56	2863.21	-1084.64	-1084.64
40	50	1688.56	2863.21	-1084.64	-1084.64
50	60	1688.56	2863.21	-1084.64	-1084.64
60	70	1688.56	2863.21	-1084.64	-1084.64
70	80	1688.56	2863.21	-1084.64	-1084.64
80	90	1688.56	2863.21	-1084.64	-1084.64
90	100	1688.56	2863.21	-1084.64	-1084.64
100	110	1688.56	2863.21	-1084.64	-1084.64
110	120	1688.56	2863.21	-1084.64	-1084.64
120	130	1688.56	2863.21	-1084.64	-1084.64



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	675.786	...	906.431
40	50	675.786	...	906.431
50	60	675.786	...	906.431
60	70	675.786	...	906.431
70	80	675.786	...	906.431
80	90	675.786	...	906.431
90	100	675.786	...	906.431
100	110	675.786	...	906.431
110	120	675.786	...	906.431
120	130	675.786	...	906.431

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-82.9997	-82.9997	-355.706	18.0867	18.0867
20	30	-79.2385	-79.2385	-351.945	18.0867	18.0867
30	40	-38.3659	-38.3659	-38.3659	8.97748	8.97748
40	50	-36.1853	-36.1853	-36.1853	8.97748	8.97748
50	60	-29.8294	-29.8294	-29.8294	8.97748	8.97748
60	70	-25.7967	-25.7967	-25.7967	7.18198	7.18198
70	80	-21.9786	-21.9786	-21.9786	5.38649	5.38649
80	90	-18.1606	-18.1606	-18.1606	3.59099	3.59099
90	100	-15.3191	-15.3191	-15.3191	3.59099	3.59099
100	110	-11.5011	-11.5011	-11.5011	1.7955	1.7955
110	120	-7.68301	-7.68301	-7.68301	...	...
120	130	-5.23525	-5.23525	-5.23525	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	18.0867	265.026	169.75	87.4585	...
20	30	18.0867	229.342	152.33	75.6829	...
30	40	8.97748	99.3645	68.2238	32.7903	...
40	50	8.97748	98.9833	68.0258	32.6645	...
50	60	8.97748	77.3408	56.3579	25.5225	...
60	70	7.18198	58.1734	45.2118	19.1972	...
70	80	5.38649	41.6089	34.6708	13.7309	...
80	90	3.59099	27.8789	24.9346	9.20003	...
90	100	3.59099	16.6949	16.1279	5.50932	...
100	110	1.7955	8.06933	8.51717	2.66288	...
110	120	...	2.51498	2.63582	0.82994	...
120	130	...	0.44683	0.027863	0.14745	...

Longitudinal Stresses due to these Conditions:



From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	169.75
20	30	...	...	...	152.33
30	40	...	...	...	68.2238
40	50	...	...	...	68.0258
50	60	...	...	...	56.3579
60	70	...	...	...	45.2118
70	80	...	...	...	34.6708
80	90	...	...	...	24.9346
90	100	...	...	...	16.1279
100	110	...	...	...	8.51717
110	120	...	...	...	2.63582
120	130	...	...	...	0.027863

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	...	-4.7806
20	30	...	-4.74277
30	40	...	-2.30311
40	50	...	-2.17644
50	60	...	-1.96923
60	70	...	-1.77739
70	80	...	-1.55955
80	90	...	-1.34452
90	100	...	-1.10611
100	110	...	-0.80881
110	120	...	-0.54935
120	130	...	-0.22266

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...
120	130	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...



60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	200.11	1181.99	-366.11	678.56	0.1693	0.5395
20	168.19	1350.85	-326.67	678.56	0.1245	0.4814
30	69.98	1688.56	-146.71	1084.64	0.0414	0.1353
40	71.78	1688.56	-144.15	1084.64	0.0425	0.1329
50	56.49	1688.56	-116.15	1084.64	0.0335	0.1071
60	39.56	1688.56	-91.15	1084.64	0.0234	0.0840
70	25.02	1688.56	-68.97	1084.64	0.0148	0.0636
80	13.31	1688.56	-49.63	1084.64	0.0079	0.0458
90	4.97	1688.56	-35.61	1084.64	0.0029	0.0328
100		1688.56	-21.37	1084.64		0.0197
110		1688.56	-10.20	1084.64		0.0094
120		1688.56	-5.68	1084.64		0.0052

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	100.06	1181.99	-275.62	678.56	0.0847	0.4062
20	86.44	1350.85	-254.40	678.56	0.0640	0.3749
30	36.53	1688.56	-117.87	1084.64	0.0216	0.1087
40	38.64	1688.56	-115.37	1084.64	0.0229	0.1064
50	33.54	1688.56	-97.13	1084.64	0.0199	0.0896
60	24.82	1688.56	-79.97	1084.64	0.0147	0.0737
70	16.52	1688.56	-63.60	1084.64	0.0098	0.0586



80	9.02	1688.56	-48.03	1084.64	0.0053	0.0443
90	3.29	1688.56	-36.14	1084.64	0.0020	0.0333
100		1688.56	-22.62	1084.64		0.0209
110		1688.56	-10.87	1084.64		0.0100
120		1688.56	-5.49	1084.64		0.0051

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	200.11	1181.99	-366.11	678.56	0.1693	0.5395
20	168.19	1350.85	-326.67	678.56	0.1245	0.4814
30	69.98	1688.56	-146.71	1084.64	0.0414	0.1353
40	71.78	1688.56	-144.15	1084.64	0.0425	0.1329
50	56.49	1688.56	-116.15	1084.64	0.0335	0.1071
60	39.56	1688.56	-91.15	1084.64	0.0234	0.0840
70	25.02	1688.56	-68.97	1084.64	0.0148	0.0636
80	13.31	1688.56	-49.63	1084.64	0.0079	0.0458
90	4.97	1688.56	-35.61	1084.64	0.0029	0.0328
100		1688.56	-21.37	1084.64		0.0197
110		1688.56	-10.20	1084.64		0.0094
120		1688.56	-5.68	1084.64		0.0052

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	100.06	1181.99	-275.62	678.56	0.0847	0.4062
20	86.44	1350.85	-254.40	678.56	0.0640	0.3749
30	36.53	1688.56	-117.87	1084.64	0.0216	0.1087
40	38.64	1688.56	-115.37	1084.64	0.0229	0.1064
50	33.54	1688.56	-97.13	1084.64	0.0199	0.0896
60	24.82	1688.56	-79.97	1084.64	0.0147	0.0737
70	16.52	1688.56	-63.60	1084.64	0.0098	0.0586
80	9.02	1688.56	-48.03	1084.64	0.0053	0.0443
90	3.29	1688.56	-36.14	1084.64	0.0020	0.0333
100		1688.56	-22.62	1084.64		0.0209
110		1688.56	-10.87	1084.64		0.0100
120		1688.56	-5.49	1084.64		0.0051

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-461.25	678.56		0.6797
20		2290.57	-445.71	678.56		0.6569
30	3.40	2863.21	-80.13	1084.64	0.0012	0.0739
40	5.46	2863.21	-77.83	1084.64	0.0019	0.0718
50	4.67	2863.21	-64.33	1084.64	0.0016	0.0593
60	0.58	2863.21	-52.18	1084.64	0.0002	0.0481
70		2863.21	-41.10	1084.64		0.0379
80		2863.21	-30.95	1084.64		0.0285
90		2863.21	-24.42	1084.64		0.0225
100		2863.21	-15.96	1084.64		0.0147
110		2863.21	-8.51	1084.64		0.0078
120		2863.21	-5.38	1084.64		0.0050

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-373.79	678.56		0.5509
20		2290.57	-370.03	678.56		0.5453
30		2863.21	-47.34	1084.64		0.0436
40		2863.21	-45.16	1084.64		0.0416
50		2863.21	-38.81	1084.64		0.0358
60		2863.21	-32.98	1084.64		0.0304
70		2863.21	-27.37	1084.64		0.0252
80		2863.21	-21.75	1084.64		0.0201
90		2863.21	-18.91	1084.64		0.0174



100	2863.21	-13.30	1084.64	0.0123
110	2863.21	-7.68	1084.64	0.0071
120	2863.21	-5.24	1084.64	0.0048

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	200.11	1181.99	-366.11	678.56	0.1693	0.5395
20	168.19	1350.85	-326.67	678.56	0.1245	0.4814
30	745.76	1688.56		1084.64	0.4417	
40	747.56	1688.56		1084.64	0.4427	
50	732.28	1688.56		1084.64	0.4337	
60	715.34	1688.56		1084.64	0.4236	
70	700.80	1688.56		1084.64	0.4150	
80	689.10	1688.56		1084.64	0.4081	
90	680.75	1688.56		1084.64	0.4032	
100	674.15	1688.56		1084.64	0.3992	
110	670.62	1688.56		1084.64	0.3972	
120	671.00	1688.56		1084.64	0.3974	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	100.06	1181.99	-275.62	678.56	0.0847	0.4062
20	86.44	1350.85	-254.40	678.56	0.0640	0.3749
30	712.32	1688.56		1084.64	0.4218	
40	714.43	1688.56		1084.64	0.4231	
50	709.32	1688.56		1084.64	0.4201	
60	700.61	1688.56		1084.64	0.4149	
70	692.31	1688.56		1084.64	0.4100	
80	684.81	1688.56		1084.64	0.4056	
90	679.08	1688.56		1084.64	0.4022	
100	673.79	1688.56		1084.64	0.3990	
110	670.19	1688.56		1084.64	0.3969	
120	670.36	1688.56		1084.64	0.3970	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	200.11	1181.99	-366.11	678.56	0.1693	0.5395
20	168.19	1350.85	-326.67	678.56	0.1245	0.4814
30	69.98	1688.56	-146.71	1084.64	0.0414	0.1353
40	71.78	1688.56	-144.15	1084.64	0.0425	0.1329
50	56.49	1688.56	-116.15	1084.64	0.0335	0.1071
60	39.56	1688.56	-91.15	1084.64	0.0234	0.0840
70	25.02	1688.56	-68.97	1084.64	0.0148	0.0636
80	13.31	1688.56	-49.63	1084.64	0.0079	0.0458
90	4.97	1688.56	-35.61	1084.64	0.0029	0.0328
100		1688.56	-21.37	1084.64		0.0197
110		1688.56	-10.20	1084.64		0.0094
120		1688.56	-5.68	1084.64		0.0052

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	100.06	1181.99	-275.62	678.56	0.0847	0.4062
20	86.44	1350.85	-254.40	678.56	0.0640	0.3749
30	36.53	1688.56	-117.87	1084.64	0.0216	0.1087
40	38.64	1688.56	-115.37	1084.64	0.0229	0.1064
50	33.54	1688.56	-97.13	1084.64	0.0199	0.0896
60	24.82	1688.56	-79.97	1084.64	0.0147	0.0737
70	16.52	1688.56	-63.60	1084.64	0.0098	0.0586
80	9.02	1688.56	-48.03	1084.64	0.0053	0.0443
90	3.29	1688.56	-36.14	1084.64	0.0020	0.0333
100		1688.56	-22.62	1084.64		0.0209
110		1688.56	-10.87	1084.64		0.0100



120 1688.56 -5.49 1084.64 0.0051

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-461.25	678.56		0.6797
20		2290.57	-445.71	678.56		0.6569
30	909.83	2863.21		1084.64	0.3178	
40	911.89	2863.21		1084.64	0.3185	
50	911.10	2863.21		1084.64	0.3182	
60	907.01	2863.21		1084.64	0.3168	
70	903.57	2863.21		1084.64	0.3156	
80	901.06	2863.21		1084.64	0.3147	
90	900.21	2863.21		1084.64	0.3144	
100	899.39	2863.21		1084.64	0.3141	
110	899.58	2863.21		1084.64	0.3142	
120	901.34	2863.21		1084.64	0.3148	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-373.79	678.56		0.5509
20		2290.57	-370.03	678.56		0.5453
30	877.04	2863.21		1084.64	0.3063	
40	879.22	2863.21		1084.64	0.3071	
50	885.58	2863.21		1084.64	0.3093	
60	887.82	2863.21		1084.64	0.3101	
70	889.84	2863.21		1084.64	0.3108	
80	891.86	2863.21		1084.64	0.3115	
90	894.70	2863.21		1084.64	0.3125	
100	896.73	2863.21		1084.64	0.3132	
110	898.75	2863.21		1084.64	0.3139	
120	901.20	2863.21		1084.64	0.3147	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-101.09	678.56		0.1490
20		1350.85	-97.33	678.56		0.1434
30	646.40	1688.56		1084.64	0.3828	
40	648.58	1688.56		1084.64	0.3841	
50	654.93	1688.56		1084.64	0.3879	
60	657.17	1688.56		1084.64	0.3892	
70	659.19	1688.56		1084.64	0.3904	
80	661.22	1688.56		1084.64	0.3916	
90	664.06	1688.56		1084.64	0.3933	
100	666.08	1688.56		1084.64	0.3945	
110	668.10	1688.56		1084.64	0.3957	
120	670.55	1688.56		1084.64	0.3971	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-71.93	678.56		0.1060
20		1350.85	-68.67	678.56		0.1012
30	642.19	1688.56		1084.64	0.3803	
40	644.10	1688.56		1084.64	0.3815	
50	649.67	1688.56		1084.64	0.3847	
60	653.20	1688.56		1084.64	0.3868	
70	656.54	1688.56		1084.64	0.3888	
80	659.89	1688.56		1084.64	0.3908	
90	662.37	1688.56		1084.64	0.3923	
100	665.72	1688.56		1084.64	0.3943	
110	669.06	1688.56		1084.64	0.3962	
120	671.20	1688.56		1084.64	0.3975	



Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-101.09	678.56		0.1490
20		1350.85	-97.33	678.56		0.1434
30	646.40	1688.56		1084.64	0.3828	
40	648.58	1688.56		1084.64	0.3841	
50	654.93	1688.56		1084.64	0.3879	
60	657.17	1688.56		1084.64	0.3892	
70	659.19	1688.56		1084.64	0.3904	
80	661.22	1688.56		1084.64	0.3916	
90	664.06	1688.56		1084.64	0.3933	
100	666.08	1688.56		1084.64	0.3945	
110	668.10	1688.56		1084.64	0.3957	
120	670.55	1688.56		1084.64	0.3971	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-101.09	678.56		0.1490
20		1350.85	-97.33	678.56		0.1434
30	646.40	1688.56		1084.64	0.3828	
40	648.58	1688.56		1084.64	0.3841	
50	654.93	1688.56		1084.64	0.3879	
60	657.17	1688.56		1084.64	0.3892	
70	659.19	1688.56		1084.64	0.3904	
80	661.22	1688.56		1084.64	0.3916	
90	664.06	1688.56		1084.64	0.3933	
100	666.08	1688.56		1084.64	0.3945	
110	668.10	1688.56		1084.64	0.3957	
120	670.55	1688.56		1084.64	0.3971	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-101.09	678.56		0.1490
20		1350.85	-97.33	678.56		0.1434
30		1688.56	-47.34	1084.64		0.0436
40		1688.56	-45.16	1084.64		0.0416
50		1688.56	-38.81	1084.64		0.0358
60		1688.56	-32.98	1084.64		0.0304
70		1688.56	-27.37	1084.64		0.0252
80		1688.56	-21.75	1084.64		0.0201
90		1688.56	-18.91	1084.64		0.0174
100		1688.56	-13.30	1084.64		0.0123
110		1688.56	-7.68	1084.64		0.0071
120		1688.56	-5.24	1084.64		0.0048

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		984.99	-105.87	565.47		0.1872
20		1125.71	-102.07	565.47		0.1805
30	644.09	1407.13		903.87	0.4577	
40	646.40	1407.13		903.87	0.4594	
50	652.97	1407.13		903.87	0.4640	
60	655.39	1407.13		903.87	0.4658	
70	657.63	1407.13		903.87	0.4674	
80	659.87	1407.13		903.87	0.4689	
90	662.95	1407.13		903.87	0.4711	
100	665.27	1407.13		903.87	0.4728	
110	667.55	1407.13		903.87	0.4744	
120	670.33	1407.13		903.87	0.4764	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10		984.99	-105.87	565.47		0.1872
20		1125.71	-102.07	565.47		0.1805
30		1407.13	-49.65	903.87		0.0549
40		1407.13	-47.34	903.87		0.0524
50		1407.13	-40.78	903.87		0.0451
60		1407.13	-34.76	903.87		0.0385
70		1407.13	-28.92	903.87		0.0320
80		1407.13	-23.10	903.87		0.0256
90		1407.13	-20.02	903.87		0.0221
100		1407.13	-14.11	903.87		0.0156
110		1407.13	-8.23	903.87		0.0091
120		1407.13	-5.46	903.87		0.0060

Absolute Maximum of the all of the Stress Ratio's 0.6797

Governing Element: headBottom
 Governing Load Case 5 : NP+HW+HI

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
headBottom	...	-355.7	18.1	87.5	...	+ Tens
headBottom	...	-355.7	-18.1	-87.5	-461.3	- Comp
Skirt-1	...	-351.9	18.1	75.7	...	+ Tens
Skirt-1	...	-351.9	-18.1	-75.7	-445.7	- Comp
Bottom head	...	-38.4	9.0	32.8	3.4	+ Tens
Bottom head	...	-38.4	-9.0	-32.8	-80.1	- Comp
Shell-1	...	-36.2	9.0	32.7	5.5	+ Tens
Shell-1	...	-36.2	-9.0	-32.7	-77.8	- Comp
Shell-2	...	-29.8	9.0	25.5	4.7	+ Tens
Shell-2	...	-29.8	-9.0	-25.5	-64.3	- Comp
Shell-3	...	-25.8	7.2	19.2	0.6	+ Tens
Shell-3	...	-25.8	-7.2	-19.2	-52.2	- Comp
Shell-4	...	-22.0	5.4	13.7	...	+ Tens
Shell-4	...	-22.0	-5.4	-13.7	-41.1	- Comp
Shell-5	...	-18.2	3.6	9.2	...	+ Tens
Shell-5	...	-18.2	-3.6	-9.2	-31.0	- Comp
Shell-6	...	-15.3	3.6	5.5	...	+ Tens
Shell-6	...	-15.3	-3.6	-5.5	-24.4	- Comp
Shell-7	...	-11.5	1.8	2.7	...	+ Tens
Shell-7	...	-11.5	-1.8	-2.7	-16.0	- Comp
Shell-8	...	-7.7	...	0.8	...	+ Tens
Shell-8	...	-7.7	...	-0.8	-8.5	- Comp
Top head	...	-5.2	...	0.1	...	+ Tens
Top head	...	-5.2	...	-0.1	-5.4	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

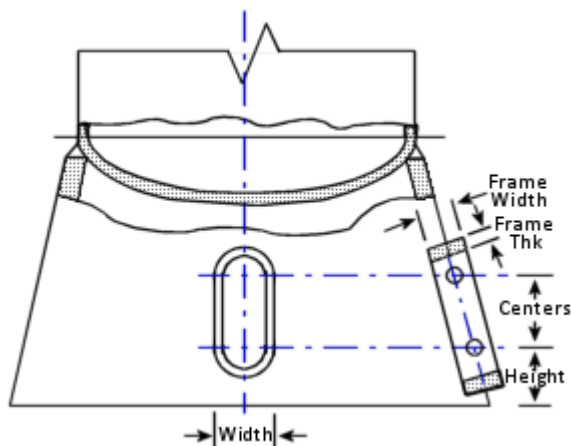
Total vessel (empty) weight	Fg	119914.953 kgf
Head Weight below section 2-2	DeltaFg	6728.364 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	119914.953 kgf
Skirt Mean Diameter	Dz	4677.000 mm
Skirt Analysis Thickness (Corroded)	ez	10.400 mm
Skirt Height	H	1850.000 mm
Additional Axial Force at section 1-1	F1	0.000 kgf
Moment at section 1-1	M1	356894.062 kgf-m.
Applied Moment - Bottom of Skirt	Mb	408871.094 kgf-m.
Weight at Bottom of Skirt	W	121085.734 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	4677.000 mm
Head Corroded Thickness	eb	21.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1165.5 \text{ mm} \\
 K &= D_i / (2 * h_i) = 4656.0 / (2 * 1165.5) = 1.9974 \\
 R &= D_i (0.44 * K + 0.02) = 4656.0 * (0.44 * 1.997 + 0.02) = 4185.1265 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 4656.0 * ((0.5 / 1.997) - 0.08) = 793.02 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	41782556.000 cm ⁴
Moment of Inertia	Iyy =	41782556.000 cm ⁴
Section Modulus	W =	178672624.000 mm ³
Section Metal Area	A =	1528.096 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(41782556, 41782556) / 1528.096)} \\
 &= 1653.569 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:



$$\begin{aligned} &= rg / L \\ &= 1653.569/1850.0 \\ &= 0.894 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 356894.031 kgf-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -356894.031 / 71469E+09 - 121085.734 / 1528.09619 \\ &= -129.177 \text{ kgf/cm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +356894.031 / 71469E+09 - 121085.734 / 1528.09619 \\ &= -29.303 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F1 - Fg - Ff + 4 \cdot M1 / Dz \\ &= 0.0 - 119914.953 - 0.0 + 4 \cdot 356894.031 / 4677.0 \\ &= 185318.0 \text{ kgf} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F1 - Fg - Ff - 4 \cdot M1 / Dz \\ &= 0.0 - 119914.953 - 0.0 - 4 \cdot 356894.031 / 4677.0 \\ &= -425147.9 \text{ kgf} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (185317.984 + 6728.364 + 0.0) / (\pi \cdot 4677.0 \cdot 21.0) + 12.236 \cdot 4677.0 / (4 \cdot 21.0) \\ &= 743.53 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.4E+06 + 6728.364 + 0.0) / (\pi \cdot 4677.0 \cdot 21.0) + 12.236 \cdot 4677.0 / (4 \cdot 21.0) \\ &= 545.69 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (0.0 + 6728.364) / (\pi \cdot 4677.0 \cdot 21.0) + 12.236 \cdot 4677.0 / (4 \cdot 21.0) \\ &= 683.47 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= Fzp / (\pi \cdot Dz \cdot ez) \\ &= 185317.984 / (\pi \cdot 4677.0 \cdot 10.4) \\ &= 121.27 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= Fzq / (\pi \cdot Dz \cdot ez) \\ &= -425147.906 / (\pi \cdot 4677.0 \cdot 10.4) \\ &= -278.22 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$\begin{aligned} &= 1 - 0.5(Db + eb - Dz + ez) / (r + eb) \\ &= 1 - 0.5(4677.0 + 21.0 - 4677.0 + 10.4) / (793.02 + 21.0) \\ &= 0.9807 \end{aligned}$$



$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.981) = 11.271 \text{ deg}$$

Dimension [a]:

$$\begin{aligned} &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(21.0^2 + 10.4^2 + 2 \cdot 21.0 \cdot 10.4 \cdot 0.981)} \\ &= 15.633 \text{ mm} \end{aligned}$$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 15.63 \cdot 185318.0 = 2897.04 \text{ kgf-m.} \\ M_q &= a \cdot F_{zq} = 15.63 \cdot -425147.9 = -6646.25 \text{ kgf-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{21.0}{10.4} \right)^2 \\ &= 0.398 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.398 \cdot 6 \cdot 2897.039 / (\pi \cdot 4677.0 \cdot 21.0^2) \\ &= 106.66 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.398 \cdot 6 \cdot -6646.253 / (\pi \cdot 4677.0 \cdot 21.0^2) \\ &= -244.69 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.398 \cdot 6 \cdot 2897.039 / (\pi \cdot 4677.0 \cdot 10.4^2) \\ &= 434.87 \text{ kgf/cm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.398 \cdot 6 \cdot -6646.253 / (\pi \cdot 4677.0 \cdot 10.4^2) \\ &= -997.66 \text{ kgf/cm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/D_b &= 0.827/184.134 = 0.0045 \\ y &= 125 \cdot eb/D_b = 125 \cdot 21.0/4677.0 = 0.5613 \end{aligned}$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$\begin{aligned} &= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994 \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.561 + 3.994 = 4.1351 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$\begin{aligned} &= (P + Ph) \cdot D_b \left(\left(\frac{y}{yA} \right) \cdot \alpha - 1 \right) / (4 \cdot eb) \\ &= (12.236 + 0.0) \cdot 4677.0 \left(\left(\frac{11.27}{40.0} \right) \cdot 4.135 - 1 \right) / (4 \cdot 21.0) \\ &= 112.54 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 743.533 - 106.657 + 112.536 \\ &= 749.41 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 743.533 + 106.657 - 112.536 \\ &= 737.65 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$



$$= 545.687 - 244.688 + 112.536$$

$$= 902.91 \text{ kgf/cm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 545.687 + -244.688 - 112.536$$

$$= 188.46 \text{ kgf/cm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 683.473 + 106.657 + 112.536$$

$$= 902.67 \text{ kgf/cm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 683.473 - 106.657 - 112.536$$

$$= 464.28 \text{ kgf/cm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 683.473 + -244.688 + 112.536$$

$$= 551.32 \text{ kgf/cm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 683.473 - -244.688 - 112.536$$

$$= 815.63 \text{ kgf/cm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 121.274 - 434.872$$

$$= -313.60 \text{ kgf/cm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 121.274 + 434.872$$

$$= 556.15 \text{ kgf/cm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -278.221 - -997.664$$

$$= 719.44 \text{ kgf/cm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -278.221 + -997.664$$

$$= -1275.89 \text{ kgf/cm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 1407.134(3 - (1/1.5)) \cdot (743.533/1407.134)^2$$

$$= 3959.48 \text{ kgf/cm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 1407.134(3 - (1/1.5)) \cdot (545.687/1407.134)^2$$

$$= 4080.32 \text{ kgf/cm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 1407.134(3 - (1/1.5)) \cdot (683.473/1407.134)^2$$

$$= 4000.08 \text{ kgf/cm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 1407.134(3 - (1/1.5)) \cdot (683.473/1407.134)^2$$

$$= 4000.08 \text{ kgf/cm}^2$$



For $\sigma_{3pi\text{tot}}$ & $\sigma_{3po\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 1407.134(3 - (1/1.5) \cdot (121.274/1407.134)^2)$$

$$= 4214.43 \text{ kgf/cm}^2$$

For $\sigma_{3qi\text{tot}}$ & $\sigma_{3qo\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 1407.134(3 - (1/1.5) \cdot (-278.221/1407.134)^2)$$

$$= 4184.73 \text{ kgf/cm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	749.41	3959.48	Pass	46
Stot1po	737.65	3959.48	Pass	47
Stot1qi	902.91	4080.32	Pass	48
Stot1qo	188.46	4080.32	Pass	49
Stot2pi	902.67	4000.08	Pass	50
Stot2po	464.28	4000.08	Pass	51
Stot2qi	551.32	4000.08	Pass	52
Stot2qo	815.63	4000.08	Pass	53
Stot3pi	-313.60	4214.43	Pass	54
Stot3po	556.15	4214.43	Pass	55
Stot3qi	719.44	4184.73	Pass	56
Stot3qo	-1275.89	4184.73	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	1724.513 cm
Center of Gravity of the Nozzles	1151.539 cm
Center of Gravity of the Added Weights (Operating)	1145.842 cm
Center of Gravity of the Added Weights (Empty)	1145.842 cm
Center of Gravity of Bare Shell New and Cold	1434.459 cm
Center of Gravity of Bare Shell Corroded	1432.458 cm
Vessel CG in the Operating Condition	1481.995 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	1479.298 cm
Vessel CG in the Test Condition	1533.503 cm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	4689.0000	mm
Skirt Thickness	STHK	12.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	26.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1407.13	kgf/cm²
Basing Yield Stress		2671.52	kgf/cm²
Inside Diameter of Basing	DI, D4	4487.0000	mm
Outside Diameter of Basing	DOU, D5	4893.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm
Bolt Corrosion Allowance	BCA	1.6000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	1400.02	kgf/cm²
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	4809.0000	mm
Bolt Allowable Shear Stress		1099.960	kgf/cm²
Ultimate Comp. Strength of Concrete	FPC	356.9	kgf/cm²
Allowable Comp. Strength of Concrete	FC	89.2	kgf/cm²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	159.0000	mm
Width of Gussets at Base Plate	BWDT	159.0000	mm
Gusset Plate Elastic Modulus	E	2057165.4	kgf/cm²
Gusset Plate Yield Stress	SY	2671.5	kgf/cm²
Height of Chair	h1	280.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	159.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	48.0000	mm
External Corrosion Allowance	CA	3.2000	mm
Dead Weight of Vessel	DW	126833.3	kgf
Operating Weight of Vessel	ROW	126833.3	kgf
Earthquake Moment on Basing	EQMOM	304430.8	kgf-m.
Wind Moment on Basing	WIMOM	474287.5	kgf-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 304431/4809.0) - 126833) / 16$$

$$= 7899.0 \text{ kgf}$$



Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 474288/4809.0) - 126833) / 16$$

$$= 16729.1 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/4809.0) - 126833) / 16$$

$$= -7927.1 \text{ kgf}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded)	Abss :	8.4527	cm ²
Area Available in all the Bolts	Abss * RN :	135.2439	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.173	0.259	1.129	2.721	0.465	0.773	51314.8	178148.1	5652.3
4	0.216	0.238	1.268	2.626	0.456	0.777	52584.9	179418.2	6003.1
6	0.227	0.221	1.301	2.602	0.453	0.777	52910.2	179743.5	6095.3
8	0.224	0.223	1.293	2.608	0.454	0.777	52828.6	179661.9	6072.1
10	0.223	0.223	1.291	2.609	0.454	0.777	52808.2	179641.5	6066.3
11	0.223	0.223	1.290	2.610	0.454	0.777	52798.1	179631.4	6063.4

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 52798.1 / (0.895 \cdot 4809.0 \cdot 2.61)$$

$$= 939.832 \text{ kgf/cm}^2, \text{ Should be } \leq 1400.0 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 16 \cdot 8.453 / (\pi \cdot 4809.0)$$

$$= 0.895 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 27.463[(2 \cdot 0.2231 \cdot 4809.0 + 203.0) / (2 \cdot 0.2231 \cdot 4809.0)]$$

$$= 30.062 \text{ kgf/cm}^2, \text{ Should be } \leq 89.2 \text{ kgf/cm}^2$$

Values for table 10.3, l = 102.0, b = 472.122, l/b = 0.216046

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(338.755, 1417.846) = 1417.846 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1417.846 / 1763.2)} + 3.2$$

$$= 25.165 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 159.0, b = 102.0, b/l = 0.641509

As b/l (0.642) is less than 1, inverting b/l = 1.559.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 11834.01 / (4 \cdot \pi) [1.3(\ln((2 \cdot 159.0 \sin(\pi \cdot 60.0 / 159.0)) / (\pi \cdot 35.0))) + 1] - [(0.7 - 0.064) \cdot 11834.01 / (4 \cdot \pi)]$$

$$= 1549.8403 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot \text{Abs}(1549.84) / 1781.01)} + 3.2$$



= 26.0500 mm

Required Gusset Plate Thickness [tg]:

$$\begin{aligned} &= P / (\text{Stress Term} \cdot l_{avg}) + C_a \\ &= 11834.01 / (1265.526 \cdot 159.0) + 3.2 \\ &= 12.725 \text{ (not less than } 9.525 + 3.2 \text{) mm} \end{aligned}$$

Bolt spacing [m]:

$$\begin{aligned} &= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts} \\ &= \pi \cdot 4809.0 / 16 = 944.245 \text{ mm} \end{aligned}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$\begin{aligned} &= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a \\ &= 1.76(11834 \cdot 60.0 / (944.24 \cdot 242.0 \cdot 2111))^{2/3} \cdot 2344.51^{1/3} + C_a \\ &= 6.519 + 1.6 = 8.119 \text{ mm} \end{aligned}$$

Shear Stress in a Single Bolt [tb]:

$$\begin{aligned} &= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 30391.6 / (8.453 \cdot 16) \\ &= 224.7 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2. \end{aligned}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	25.1654	mm
Actual Basing Thickness as entered by user	26.0000	mm
Required Thickness of Chair Cap	26.0500	mm
Actual Top Ring Thickness as entered by user	38.0000	mm
Required Gusset thickness, + CA	12.7250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	8.1188	mm
Given Thickness of Skirt	12.0000	mm
Required Gusset Height to meet local stress	128.2923	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$\begin{aligned} &= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A) \\ &= 461.252 \cdot (12.0 - 1.6) \\ &= 47.97 \text{ kgf/mm} \end{aligned}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$\begin{aligned} &= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 48 / [(0.4 \cdot 2508) \cdot 2 \cdot 0.707] \\ &= 3.381 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (C_{mwth} + 2 \cdot (H_g + T_{ta})) \\ &= 11834.0 / (174.0 + 2 \cdot (216.0 + 38.0)) \\ &= 17.352 \text{ kgf/mm} \end{aligned}$$

Horizontal Plate Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (C_{mwth} \cdot (H_g + T_{ta}) + 0.6667 \cdot (H_g + T_{ta})^2) \\ &= 11834.0 \cdot 60.0 / (174.0 \cdot (254.0) + 0.6667 \cdot (254.0)^2) \\ &= 8.142 \text{ kgf/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{ W_v^2 + W_h^2 } \\ &= \sqrt{ 17.35^2 + 8.14^2 } \\ &= 19.167 \text{ kgf/mm} \end{aligned}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 19.17 / [(0.4 \cdot 2508) \cdot 2 \cdot 0.707] \\ &= 1.351 \text{ mm} \end{aligned}$$



Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 11834.0 / (2 \cdot 159.0)$$

$$= 37.214 \text{ kgf/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 11834.0 \cdot 60.0 / (2 \cdot 254.0 \cdot 159.0)$$

$$= 8.791 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{37.21^2 + 8.79^2}$$

$$= 38.238 \text{ kgf/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 38.24 / [(0.4 \cdot 2508) \cdot 2 \cdot 0.707]$$

$$= 2.695 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	2.6952	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N3 100

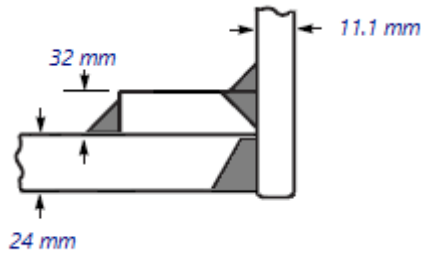
From: 30

Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Elliptical Head	D	4650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	-48.5000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1202.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1202.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.24 \cdot 4656.0 \cdot 0.8989 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
 $= 18.2129 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 49.02 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 0.5020 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm
Parallel to Vessel Wall, opening length	d	98.0496 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1202.2/1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1202.2/1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 1406.1/1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.854, 0.999)$
 $= 0.854$

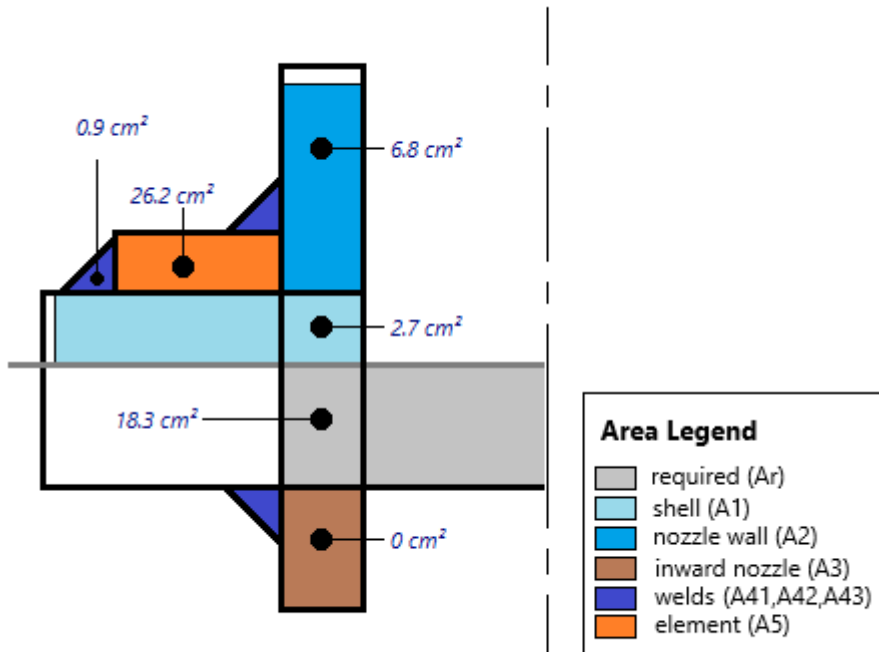


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.289	NA	NA
Area in Shell	A1	2.667	NA	NA
Area in Nozzle Wall	A2	6.814	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.854	NA	NA
Area in Element	A5	26.157	NA	NA
TOTAL AREA AVAILABLE	Atot	36.493	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 139.1717 32.0000 mm
 Based on given Pad Diameter: 214.3000 9.7299 mm
 Based on Shell or Nozzle Thickness: 185.8399 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (98.0496 \cdot 18.2129 \cdot 1.0 + 2 \cdot 8.1252 \cdot 18.2129 \cdot 1.0 \cdot (1 - 0.85))$
 $= 18.289 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 98.05(1.0 \cdot 21.0 - 1.0 \cdot 18.213) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 18.2129) \cdot (1 - 0.854)$
 $= 2.667 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.5) \cdot 0.8544$



$$= 6.814 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.85 + (0.0)^2 \cdot 0.85 + 0.0^2 \cdot 1.$
 $= 0.854 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (196.0992 - 114.3) \cdot 32.0 \cdot 0.9993$
 $= 26.157 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$ta = 3.5020 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 23.2267 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 23.2267 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(23.2267, 4.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.502, 8.2578)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 327.2,	Allowable	: 1202.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2678.3 kgf/cm ²	Passed
Occasional	: 34.1,	Allowable	: 1598.9 kgf/cm ²	Passed
Shear	: 212.3,	Allowable	: 841.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.735$, $tr = 0.502$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.075$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.735$, $tr = 0.502$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.075$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.735$, $tr = 0.502$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.075$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24.0, tr = 18.213, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.867$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-9 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.075$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $12.24/52.01 = 0.235$

Weld Size Calculations, Description: N3 100

Intermediate Calc. for nozzle/shell Welds	Tmin	8.1252 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (18.2887 - 2.6668 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 18.2129)) \cdot 1407)$
 $= 22526.61 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6.8142 + 26.1572 + 0.8544 - 0.0 \cdot 0.85) \cdot 1407$
 $= 47597.43 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6.8142 + 0.0 + 0.8544 + (2.9156)) \cdot 1407$
 $= 14893.34 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6.8142 + 0.0 + 0.8544 + 26.1572 + (2.9156)) \cdot 1407$
 $= 51700.05 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 1202$
 $= 10576. \text{ kgf}$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 214.3 \cdot 24.0 \cdot 0.49 \cdot 1406$
 $= 55664. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 53.0874) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202$
 $= 11404. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 114.3 \cdot 32.0 \cdot 0.74 \cdot 1406$
 $= 59783. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 114.3 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407$
 $= 39260. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (55664 + 11404) = 67068 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (10576 + 59783 + 39260 + 0) = 109619 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (55664 + 39260 + 0) = 94925 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 67068	kgf, must exceed W = 22526 kgf	or W1 = 47597 kgf
Path 2-2 = 109619	kgf, must exceed W = 22526 kgf	or W2 = 14893 kgf
Path 3-3 = 94924	kgf, must exceed W = 22526 kgf	or W3 = 51700 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.3883 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 274.3884 mm

Input Echo, WRC107/537 Item 1, Description: N3 100 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	8370.000	mm
Vessel Thickness	Tv	24.000	mm
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID

Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	92.050	mm
Nozzle Thickness	Tn	11.125	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	1202.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1202.20	kgf/cm ²
Thickness of Reinforcing Pad	Tpad	32.000	mm
Diameter of Reinforcing Pad	Dpad	214.300	mm
Design Internal Pressure	Dp	12.236	kgf/cm ²
Include Pressure Thrust		No	

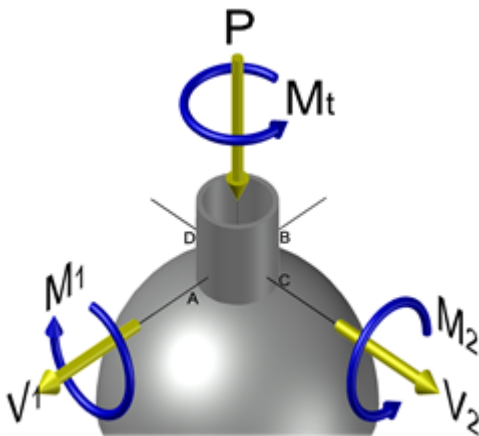
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-275.3	kgf
Longitudinal Shear (SUS)	(V1) V1	407.9	kgf
Circumferential Shear (SUS)	(Vc) V2	407.9	kgf
Circumferential Moment (SUS)	(Mc) M1	122.3	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	163.1	kgf-m.
Torsional Moment (SUS)	Mt	203.9	kgf-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{4198.5(24.0 - 3.0)}$$

$$= 1094.175 \text{ mm}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	-275.3	kgf
Circumferential Shear	(VC) V2	407.9	kgf
Longitudinal Shear	(VL) V1	407.9	kgf
Circumferential Moment	(MC) M1	122.3	kgf-m.
Longitudinal Moment	(ML) M2	163.1	kgf-m.
Torsional Moment	MT	203.9	kgf-m.

Dimensionless Param: U = 0.12 TAU = 6.53 RHO = 4.00 (6.52)



Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10221	(A,B,C,D)
$M(x) / P$	SP 4	0.04139	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / MC$	SM 4	0.44886	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / MC$	SM 4	0.20071	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / ML$	SM 4	0.44886	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / ML$	SM 4	0.20071	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33450	(A,B,C,D)
$M(y) / P$	SP 4	0.28869	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / MC$	SM 4	0.10287	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / MC$	SM 4	1.53219	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / ML$	SM 4	0.10287	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / ML$	SM 4	1.53219	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Rad. Bend. P		2.4	-2.4	2.4	-2.4	2.4	-2.4	2.4	-2.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-4.1	-4.1	4.1	4.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-11.1	11.1	11.1	-11.1
Rad. Memb. ML		-5.5	-5.5	5.5	5.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-14.8	14.8	14.8	-14.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-16.9	7.8	23.7	-10.7	-11.8	5.5	18.7	-8.4
Tang. Memb. P		3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Tang. Bend. P		17.0	-17.0	17.0	-17.0	17.0	-17.0	17.0	-17.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-84.7	84.7	84.7	-84.7
Tang. Memb. ML		-1.3	-1.3	1.3	1.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-112.9	112.9	112.9	-112.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-93.9	98.0	134.5	-125.4	-65.4	70.1	105.9	-97.5
Shear VC		4.3	4.3	-4.3	-4.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.3	-4.3	4.3	4.3
Shear MT		18.7	18.7	18.7	18.7	18.7	18.7	18.7	18.7
Tot. Shear		23.0	23.0	14.5	14.5	14.5	14.5	23.0	23.0
Str. Int.		100.3	103.5	136.3	127.2	69.0	73.1	111.6	103.1

Unitless Prm: $U = 0.36$ TAU = 0.00 (12.69) RHO = 0.00 (2.58)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.17126	(A,B,C,D)
$M(x) / P$	SR 2	0.11785	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / MC$	SR 3	0.16020	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / MC$	SR 3	0.38154	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / ML$	SR 3	0.16020	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / ML$	SR 3	0.38154	(A,B,C,D)
$N(y) * T / P$	SR 2	0.05136	(A,B,C,D)
$M(y) / P$	SR 2	0.03531	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / MC$	SR 3	0.04928	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / MC$	SR 3	0.11393	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / ML$	SR 3	0.04928	(A,B,C,D)



$M(y) * \sqrt{R_m * T} / ML \quad SR \quad 3 \quad 0.11393 \quad (A,B,C,D)$

Stress Concentration Factors: $K_n = 1.00, K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		10.7	10.7	10.7	10.7	10.7	10.7	10.7	10.7
Rad. Bend. P		44.1	-44.1	44.1	-44.1	44.1	-44.1	44.1	-44.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-15.0	-15.0	15.0	15.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-213.8	213.8	213.8	-213.8
Rad. Memb. ML		-20.0	-20.0	20.0	20.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-285.1	285.1	285.1	-285.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-250.2	231.7	359.9	-298.6	-174.0	165.4	283.6	-232.3
Tang. Memb. P		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Tang. Bend. P		13.2	-13.2	13.2	-13.2	13.2	-13.2	13.2	-13.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-4.6	-4.6	4.6	4.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-63.9	63.9	63.9	-63.9
Tang. Memb. ML		-6.1	-6.1	6.1	6.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-85.1	85.1	85.1	-85.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-74.8	69.0	107.7	-89.0	-52.0	49.2	84.9	-69.3
Shear VC		5.8	5.8	-5.8	-5.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-5.8	-5.8	5.8	5.8
Shear MT		13.5	13.5	13.5	13.5	13.5	13.5	13.5	13.5
Tot. Shear		19.2	19.2	7.7	7.7	7.7	7.7	19.2	19.2
Str. Int.		252.3	234.0	360.2	298.9	174.5	165.9	285.5	234.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		483.5	483.5	483.5	483.5	483.5	483.5	483.5	483.5
Rad. Pl (SUS)		-4.5	-4.5	6.5	6.5	-3.1	-3.1	5.1	5.1
Rad. Q (SUS)		-12.4	12.4	17.2	-17.2	-8.7	8.7	13.5	-13.5
Long. Pm (SUS)		483.5	483.5	483.5	483.5	483.5	483.5	483.5	483.5
Long. Pl (SUS)		2.0	2.0	4.5	4.5	2.3	2.3	4.2	4.2
Long. Q (SUS)		-96.0	96.0	129.9	-129.9	-67.7	67.7	101.7	-101.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.3	4.3	-4.3	-4.3	-4.3	-4.3	4.3	4.3
Shear Q (SUS)		18.7	18.7	18.7	18.7	18.7	18.7	18.7	18.7
Pm (SUS)		483.5	483.5	483.5	483.5	483.5	483.5	483.5	483.5
Pm+Pl (SUS)		487.6	487.6	493.4	493.4	488.2	488.2	492.5	492.5
Pm+Pl+Q (Total)		473.0	587.0	619.8	474.6	475.3	556.6	595.1	480.7

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	483.49	1407.13	Passed



Pm+Pl (SUS)	493.42	2110.70	Passed
Pm+Pl+Q (TOTAL)	619.81	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1
Rad.	Pl (SUS)	-9.3	-9.3	30.6	30.6	-4.3	-4.3	25.7	25.7
Rad.	Q (SUS)	-241.0	241.0	329.3	-329.3	-169.7	169.7	258.0	-258.0
Long.	Pm (SUS)	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1
Long.	Pl (SUS)	-2.9	-2.9	9.3	9.3	-1.4	-1.4	7.8	7.8
Long.	Q (SUS)	-71.9	71.9	98.4	-98.4	-50.6	50.6	77.1	-77.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	5.8	5.8	-5.8	-5.8	-5.8	-5.8	5.8	5.8
Shear	Q (SUS)	13.5	13.5	13.5	13.5	13.5	13.5	13.5	13.5
Pm (SUS)		1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1
Pm+Pl (SUS)		1220.6	1220.6	1252.2	1252.2	1223.2	1223.2	1247.5	1247.5
Pm+Pl+Q (Total)		1147.4	1454.1	1580.3	1131.4	1168.6	1386.1	1505.6	1153.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1220.13	1407.13	Passed
Pm+Pl (SUS)	1252.23	2110.70	Passed
Pm+Pl+Q (TOTAL)	1580.28	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N1 200

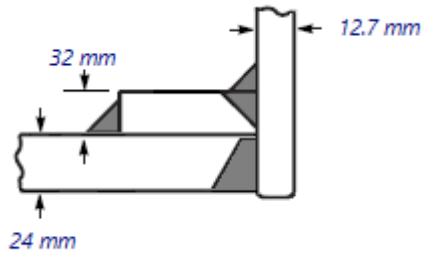
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Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4650.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		55.00	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1202.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1202.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1 200

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 2328.0 / (1407 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 20.3500 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 99.84 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 1.0224 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500	mm
Parallel to Vessel Wall, opening length	d	199.6750	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1202.2 / 1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1202.2 / 1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1406.1 / 1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.854, 0.999)$
 $= 0.854$

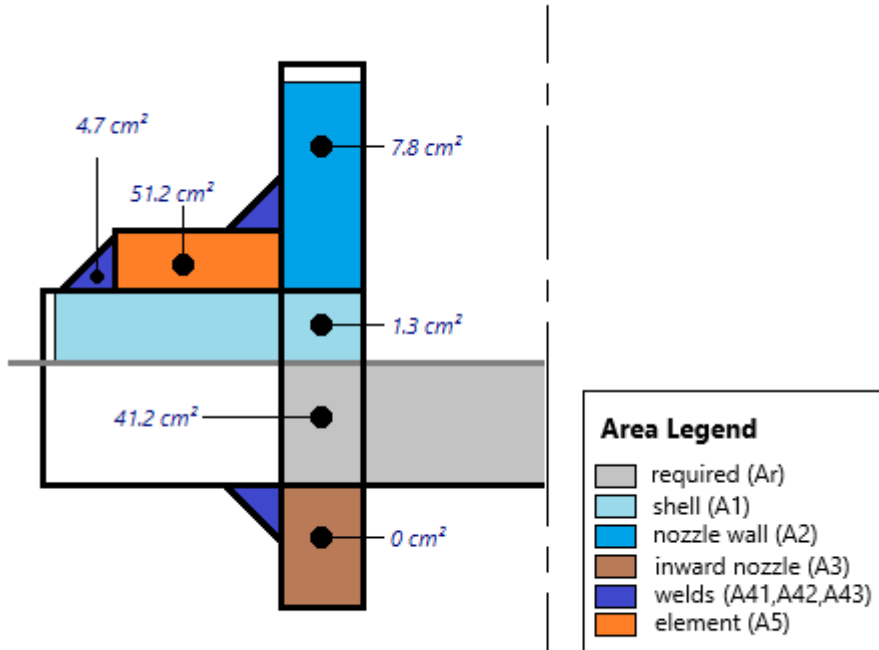


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	41.209	NA	NA
Area in Shell	A1	1.279	NA	NA
Area in Nozzle Wall	A2	7.785	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.690	NA	NA
Area in Element	A5	51.164	NA	NA
TOTAL AREA AVAILABLE	Atot	64.918	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	304.9323	32.0000 mm
Based on given Pad Diameter:	379.0750	17.1714 mm
Based on the Estimated Diameter Limit:	397.7625	15.3756 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (199.675 \cdot 20.35 \cdot 1.0 + 2 \cdot 9.7 \cdot 20.35 \cdot 1.0 \cdot (1 - 0.85))$
 $= 41.209 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199.675(1.0 \cdot 21.0 - 1.0 \cdot 20.35) - 2 \cdot 9.7$
 $(1.0 \cdot 21.0 - 1.0 \cdot 20.35) \cdot (1 - 0.854)$
 $= 1.279 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 52.5) \cdot (9.7 - 1.02) \cdot 0.8544$



$$= 7.785 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.85 + (0.0) • 0.85 + 383.8311² • 1.
= 4.690 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (379.075 - 219.075) • 32.0 • 0.9993
= 51.164 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0224 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 23.3500 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 23.3500 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[10.16, max(23.35, 4.5)]
= 10.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.0224, 10.16)
= 10.1600 mm

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 306.4,	Allowable	: 1202.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2699.1 kgf/cm ²	Passed
Occasional	: 60.1,	Allowable	: 1598.9 kgf/cm ²	Passed
Shear	: 172.2,	Allowable	: 841.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.113, tr = 1.022, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.126, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.113, tr = 1.022, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.126, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 1.022, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.126, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24.0, tr = 20.35, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.969$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -4 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 1.022, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.126$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.24 / 19.99 = 0.612$

Weld Size Calculations, Description: N1 200

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * \text{TminPad}$	$16.9680 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (41.2089 - 1.2795 + 2 \cdot 9.7 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 20.35)) \cdot 1407)$
 $= 56337.51 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.7845 + 51.1638 + 4.69 - 0.0 \cdot 0.85) \cdot 1407$
 $= 89547.51 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (7.7845 + 0.0 + 0.8544 + (3.4807)) \cdot 1407$
 $= 17053.81 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (7.7845 + 0.0 + 4.69 + 51.1638 + (3.4807)) \cdot 1407$
 $= 94445.28 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 1202$$

$$= 20272. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 379.075 \cdot 24.0 \cdot 0.49 \cdot 1406$$

$$= 98465. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 104.6875) \cdot (12.7 - 3.0) \cdot 0.7 \cdot 1202$$

$$= 26847. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 32.0 \cdot 0.74 \cdot 1406$$

$$= 114584. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407$$

$$= 75249. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (98465 + 26847) = 125311 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20272 + 114584 + 75249 + 0) = 210104 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (98465 + 75249 + 0) = 173713 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 125311 kgf, must exceed W = 56337 kgf	or W1 = 89547 kgf
Path 2-2 = 210103 kgf, must exceed W = 56337 kgf	or W2 = 17053 kgf
Path 3-3 = 173713 kgf, must exceed W = 56337 kgf	or W3 = 94445 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.5816 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.5817 mm

Input Echo, WRC297 Item 1, Description: N1 200 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4650.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm



Nozzle Thickness		12.7000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1202.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1202.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	379.075	mm
Thickness of Reinforcing Pad	32.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	12.24	kgf/cm ²
Radial Load	P	-550.67	kgf
Circumferential Shear	Vc	815.81	kgf
Longitudinal Shear	Vl	815.81	kgf
Circumferential Moment	Mc	489.29	kgf-m.
Longitudinal Moment	Ml	652.39	kgf-m.
Torsional Moment	Mt	815.48	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(2354.5(24.0 - 3.0))}$
 $= 952.867 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4709.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	53.000	mm
Nozzle Thickness used	(t) =	9.700	mm
T / t	=	5.464	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.439$$

Nr/P	=	0.105
Mr/P	=	0.076
M0/P	=	0.216
N0/P	=	0.349
MrD/Mc	=	0.098
NrDL/Mc	=	0.114
M0d/Mc	=	0.308
N0DL/Mc	=	-0.120
MrD/Ml	=	0.099
NrDL/Ml	=	0.114
M0D/Ml	=	0.292
N0DL/Ml	=	0.144

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15



Outplane Bending (ML)	-185	185	185	-185
Normal Pressure Stress	531	543	531	543

Outplane Stress Summary	362	695	764	354
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-63	63	63	-63
Inplane Pressure Stress	265	265	265	265

Inplane Stress Summary	201	309	351	207
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	20	20	20	20

Shear Stress Summary	24	24	15	15
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	366	696	764	356
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	9	9	-9	-9
Outplane Bending (Mc)	-147	147	147	-147
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	265	265	265	265

Outplane Stress Summary	160	403	435	90
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-46	46	46	-46
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	531	543	531	543

Inplane Stress Summary	486	574	598	498
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	20	20	20	20
Shear Stress Summary	15	15	24	24

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	487	576	601	500

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	144	158	144	158
Hoop Stress Summary	135	150	166	180

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	8	8	8	8
Axial Bending (P)	81	-81	81	-81
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-203	-203	203	203
Axial Bending (ML)	-804	804	804	-804
Axial Pressure Stress	71	71	71	71
Axial Stress Summary	-845	599	1170	-601

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	24	24	-24	-24
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	111	111	111	111
Shear Stress Summary	135	135	87	87

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1018	636	1177	801

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	9	9	-9	-9



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	144	158	144	158

Hoop Stress Summary	160	175	141	156
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	8	8	8	8
Axial Bending (P)	81	-81	81	-81
Axial Membrane (Mc)	-152	-152	152	152
Axial Bending (Mc)	-591	591	591	-591
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	71	71	71	71

Axial Stress Summary	-581	437	906	-439
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-24	-24	24	24
Shear due to Torsion	111	111	111	111

Shear Stress Summary	87	87	135	135
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	762	463	929	654
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(2338.5(24.0 - 3.0))}$
 $= 950.369 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	4677.000 mm
Nozzle Outside Diameter (d)	=	379.075 mm
Vessel Thickness used (T)	=	21.000 mm
Nozzle Thickness used (t)	=	89.700 mm
T / t	=	0.234
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.210$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.096
Mr/P	=	0.203
M0/P	=	0.059
N0/P	=	0.108
MrD/Mc	=	0.307
NrDL/Mc	=	0.078
M0d/Mc	=	0.094
N0DL/Mc	=	0.054
MrD/M1	=	0.169
NrDL/M1	=	0.096
M0D/M1	=	0.053



NØDL/M1 = 0.114

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	44	-44	44	-44
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-44	-44	44	44
Outplane Bending (ML)	-123	123	123	-123
Normal Pressure Stress	1350	1362	1350	1362
Outplane Stress Summary	1239	1410	1575	1253

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	151	-151	151	-151
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-37	-37	37	37
Inplane Bending (ML)	-396	396	396	-396
Inplane Pressure Stress	675	675	675	675
Inplane Stress Summary	404	894	1273	176

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	6	6	-6	-6
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	23	23	10	10

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1240	1411	1576	1253

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	44	-44	44	-44
Outplane Membrane (Mc)	-15	-15	15	15
Outplane Bending (Mc)	-164	164	164	-164
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	675	675	675	675
Outplane Stress Summary	551	793	913	495

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	151	-151	151	-151
Inplane Membrane (Mc)	-22	-22	22	22
Inplane Bending (Mc)	-539	539	539	-539
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1350	1362	1350	1362

Inplane Stress Summary	952	1739	2075	706
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-6	-6	6	6
Torsional Shear (Mt)	17	17	17	17

Shear Stress Summary	10	10	23	23
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	952	1739	2076	708

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	531	543	531	543	531	543	531	543
Circ. Pl (SUS)	-9	-9	21	21	-6	-6	10	10
Circ. Q (SUS)	-160	160	210	-210	-38	38	54	-54
Long. Pm (SUS)	265	265	265	265	265	265	265	265
Long. Pl (SUS)	-10	-10	14	14	15	15	-3	-3
Long. Q (SUS)	-55	55	71	-71	-122	122	172	-172
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	20	20	20	20	20	20	20	20
Pm (SUS)	531.0	543.0	531.0	543.0	531.0	543.0	531.0	543.0
Pm+Pl (SUS)	522.1	534.1	552.1	564.1	525.1	537.1	541.1	553.1
Pm+Pl+Q (Total)	365.5	695.5	762.6	355.7	487.8	576.5	598.5	500.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	543.00	1407.13	Passed
Pm+Pl (SUS)	564.06	2110.70	Passed
Pm+Pl+Q (TOTAL)	762.62	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1350	1362	1350	1362	1350	1362	1350	1362
Circ. Pl (SUS)	-31	-31	57	57	-11	-11	33	33
Circ. Q (SUS)	-79	79	167	-167	-388	388	690	-690
Long. Pm (SUS)	675	675	675	675	675	675	675	675
Long. Pl (SUS)	-26	-26	48	48	-2	-2	28	28
Long. Q (SUS)	-245	245	547	-547	-120	120	208	-208
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1350	1362	1350	1362	1350	1362	1350	1362
Pm+Pl (SUS)	1319	1331	1407	1419	1339	1351	1383	1395
Pm+Pl+Q (Total)	1240	1411	1574	1252	951	1739	2073	707

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1362.00	1407.13	Passed
Pm+Pl (SUS)	1419.05	2110.70	Passed
Pm+Pl+Q (TOTAL)	2073.46	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	144	158	144	158	144	158	144	158
Circ. Pl (SUS)	-9	-9	21	21	15	15	-3	-3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	71	71	71	71	71	71	71	71
Long. Pl (SUS)	-195	-195	211	211	-144	-144	160	160
Long. Q (SUS)	-723	723	885	-885	-510	510	672	-672
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	24	24	-24	-24	-24	-24	24	24
Shear Q (SUS)	111	111	111	111	111	111	111	111
Pm (SUS)	144.0	158.0	144.0	158.0	144.0	158.0	144.0	158.0
Pm+Pl (SUS)	263.4	277.2	286.7	287.3	236.9	250.6	237.0	237.9
Pm+Pl+Q (Total)	1018	636	1174	801	762	463	926	654

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	158.00	1202.20	Passed
Pm+Pl (SUS)	287.32	1803.31	Passed
Pm+Pl+Q (TOTAL)	1174.50	3606.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User
FileName : 404-V-001_R2
Nozzle Calcs.: N1 200

Nozl: 16 3:50pm Aug 11, 2026

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Input, Nozzle Desc: M1 600

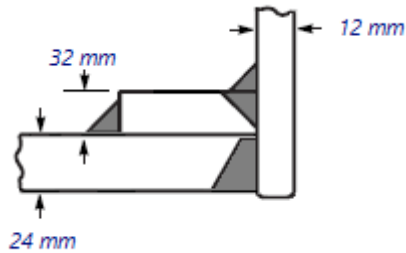
From: 50

Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4650.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		457.40	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1406.14	kgf/cm ²
Allowable Stress At Ambient	Sna	1406.14	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1273.6000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		320.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1 600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 2328.0 / (1407 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 20.3500 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 307.8 / (1406 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 2.6925 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001 mm
Parallel to Vessel Wall, opening length	d	615.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1406.1 / 1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1406.1 / 1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1406.1 / 1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.999, 0.999)$
 $= 0.999$

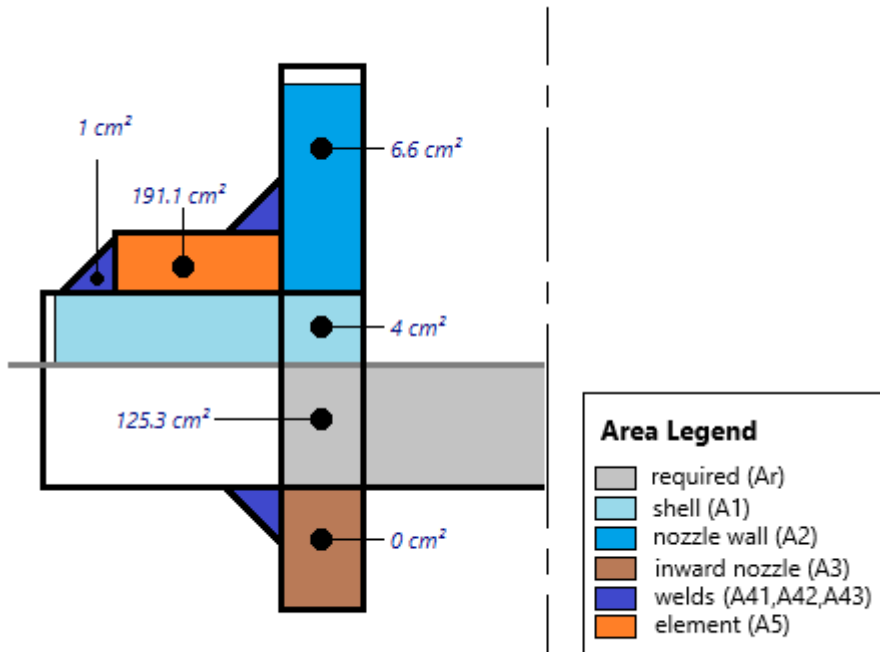


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	125.277	NA	NA
Area in Shell	A1	4.001	NA	NA
Area in Nozzle Wall	A2	6.618	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.999	NA	NA
Area in Element	A5	191.097	NA	NA
TOTAL AREA AVAILABLE	Atot	202.716	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 989.0345 32.0000 mm
 Based on given Pad Diameter: 1273.6000 19.0326 mm
 Based on the Estimated Diameter Limit: 1229.6125 19.0833 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (615.6 \cdot 20.35 \cdot 1.0 + 2 \cdot 9.0 \cdot 20.35 \cdot 1.0 \cdot (1 - 1.))$
 $= 125.277 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 615.6(1.0 \cdot 21.0 - 1.0 \cdot 20.35) - 2 \cdot 9.0$
 $(1.0 \cdot 21.0 - 1.0 \cdot 20.35) \cdot (1 - 0.999)$
 $= 4.001 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 52.5) \cdot (9.0 - 2.69) \cdot 0.9993$



= 6.618 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
= 0.999 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1231.2001 - 633.6) \cdot 32.0 \cdot 0.9993$
= 191.097 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.6925$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= t_a
= 5.6925 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 12.0$, $t_r = 2.693$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.299$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.693$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.299$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 24.0$, $t_r = 20.35$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.969$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.693$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.299$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 12.24/19.99 = 0.612

Weld Size Calculations, Description: M1 600

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	16.9680 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (125.2773 - 4.0012 + 2 • 9.0 • 0.9993 •
(1.0 • 21.0 - 20.35))1407)
= 170816.09 kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (6.6182 + 191.0969 + 0.9993 - 0.0 • 1.) • 1407
= 279617.38 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (6.6182 + 0.0 + 0.9993 + (3.7773)) • 1407
= 16033.98 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (6.6182 + 0.0 + 0.9993 + 191.0969 + (3.7773)) • 1407
= 284932.56 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 633.6 • 10.0 • 0.49 • 1406
= 68574. kgf

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.1416/2.0) • 1273.6 • 24.0 • 0.49 • 1406
= 330817. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 312.3) • (12.0 - 3.0) • 0.7 • 1406
= 86914. kgf

Tension, Pad Groove Weld [Tpgw]:
= (π/2) • Dlo • Wgpn • 0.74 • Seg
= (3.1416/2) • 633.6 • 32.0 • 0.74 • 1406
= 331394. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 633.6 • (24.0 - 3.0) • 0.74 • 1407
= 217631. kgf

See UG-41 and UW-15 for clarification of allowable stresses.



Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (330817 + 86914) = 417731 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (68574 + 331394 + 217631 + 0) = 617599 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (330817 + 217631 + 0) = 548448 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 417731 kgf, must exceed W = 170816 kgf or W1 = 279617 kgf

Path 2-2 = 617599 kgf, must exceed W = 170816 kgf or W2 = 16033 kgf

Path 3-3 = 548448 kgf, must exceed W = 170816 kgf or W3 = 284932 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 21.6843 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 295.6844 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.931 %

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Input, Nozzle Desc: N2 150

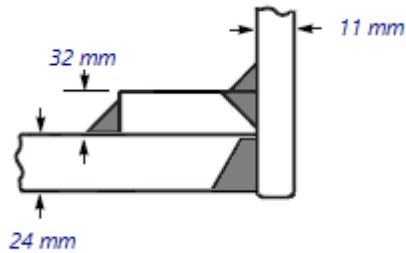
From: 120

Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Elliptical Head	D	4650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1202.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1202.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2 150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.24 \cdot 4656.0 \cdot 0.8989 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
 $= 18.2129 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 76.16 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 0.7800 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.6588 mm
Parallel to Vessel Wall, opening length	d	152.3294 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		51.9320 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1202.2/1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1202.2/1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 1406.1/1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.854, 0.999)$
 $= 0.854$

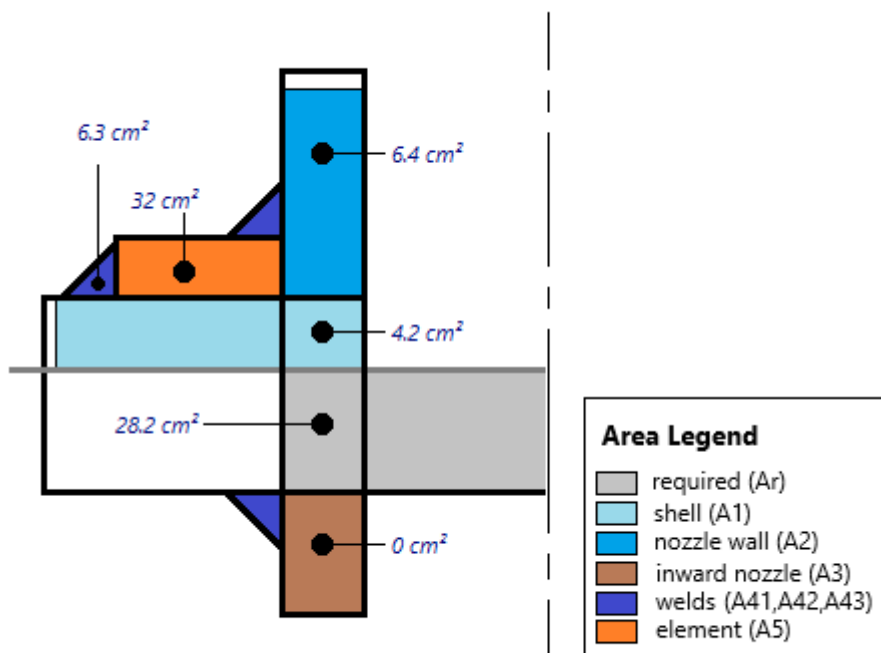


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	28.167	NA	NA
Area in Shell	A1	4.181	NA	NA
Area in Nozzle Wall	A2	6.383	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.273	NA	NA
Area in Element	A5	31.977	NA	NA
TOTAL AREA AVAILABLE	Atot	48.814	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 203.7056 32.0000 mm
 Based on given Pad Diameter: 268.2750 11.3378 mm
 Based on Shell or Nozzle Thickness: 271.6012 10.9728 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.3294 \cdot 18.2129 \cdot 1.0 + 2 \cdot 7.9728 \cdot 18.2129 \cdot 1.0 \cdot (1 - 0.85))$
 $= 28.167 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.329(1.0 \cdot 21.0 - 1.0 \cdot 18.213) - 2 \cdot 7.973$
 $(1.0 \cdot 21.0 - 1.0 \cdot 18.2129) \cdot (1 - 0.854)$
 $= 4.181 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 51.93) \cdot (7.97 - 0.78) \cdot 0.8544$



$$= 6.383 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.85 + (0.0) • 0.85 + 542.2658² • 1.
= 6.273 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (268.275 - 168.275) • 32.0 • 0.9993
= 31.977 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7800 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 23.2267 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 23.2267 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 9.2200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[9.22, max(23.2267, 4.5)]
= 9.2200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.78, 9.22)
= 9.2200 mm

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 343.9,	Allowable	: 1202.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2661.6 kgf/cm ²	Passed
Occasional	: 55.5,	Allowable	: 1598.9 kgf/cm ²	Passed
Shear	: 203.2,	Allowable	: 841.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.78, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.118, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.78, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.118, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.78, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.118, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24.0, tr = 18.213, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.867$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-9 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 0.78, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.118$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $12.24/52.01 = 0.235$

Weld Size Calculations, Description: N2 150

Intermediate Calc. for nozzle/shell Welds	Tmin	7.9728 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.5810 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (28.1665 - 4.1809 + 2 \cdot 7.9728 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 18.2129)) \cdot 1407)$
 $= 34285.33 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6.3827 + 31.9774 + 6.2732 - 0.0 \cdot 0.85) \cdot 1407$
 $= 62804.99 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6.3827 + 0.0 + 0.8544 + (2.8609)) \cdot 1407$
 $= 14209.22 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6.3827 + 0.0 + 6.2732 + 31.9774 + (2.8609)) \cdot 1407$
 $= 66830.65 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 1202$
 $= 15571. \text{ kgf}$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 268.275 \cdot 24.0 \cdot 0.49 \cdot 1406$
 $= 69684. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 80.1511) \cdot (10.9728 - 3.0) \cdot 0.7 \cdot 1202$
 $= 16895. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 32.0 \cdot 0.74 \cdot 1406$
 $= 88013. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 168.275 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407$
 $= 57800. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (69684 + 16895) = 86579 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (15571 + 88013 + 57800 + 0) = 161384 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (69684 + 57800 + 0) = 127484 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 86578	kgf, must exceed W = 34285 kgf	or W1 = 62804 kgf
Path 2-2 = 161384	kgf, must exceed W = 34285 kgf	or W2 = 14209 kgf
Path 3-3 = 127484	kgf, must exceed W = 34285 kgf	or W3 = 66830 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8422 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 524.8423 mm

Input Echo, WRC107/537 Item 1, Description: N2 150 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	8370.000	mm
Vessel Thickness	Tv	24.000	mm
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID

Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	146.329	mm
Nozzle Thickness	Tn	10.973	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	1202.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1202.20	kgf/cm ²
Thickness of Reinforcing Pad	Tpad	32.000	mm
Diameter of Reinforcing Pad	Dpad	268.275	mm
Design Internal Pressure	Dp	12.236	kgf/cm ²
Include Pressure Thrust		No	

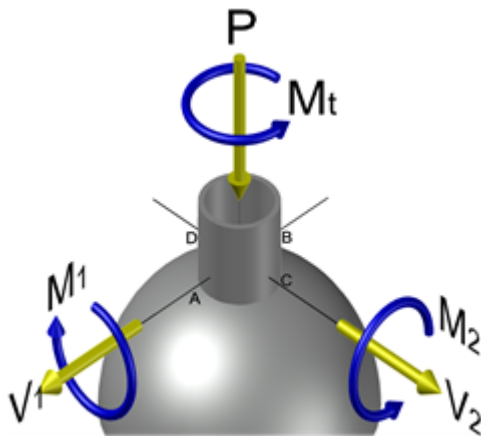
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-407.9	kgf
Longitudinal Shear (SUS)	(V1) V1	611.9	kgf
Circumferential Shear (SUS)	(Vc) V2	611.9	kgf
Circumferential Moment (SUS)	(Mc) M1	275.2	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	367.0	kgf-m.
Torsional Moment (SUS)	Mt	458.7	kgf-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(4198.5 (24.0 - 3.0))}$$

$$= 1148.150 \text{ mm}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	-407.9	kgf
Circumferential Shear	(VC) V2	611.9	kgf
Longitudinal Shear	(VL) V1	611.9	kgf
Circumferential Moment	(MC) M1	275.2	kgf-m.
Longitudinal Moment	(ML) M2	367.0	kgf-m.
Torsional Moment	MT	458.7	kgf-m.

Dimensionless Param: U = 0.18 TAU = 10.05 RHO = 4.00 (6.65)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.08042	(A,B,C,D)
$M(x) / P$	SP 4	0.03820	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / MC$	SM 4	0.29007	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / MC$	SM 4	0.15442	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / ML$	SM 4	0.29007	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / ML$	SM 4	0.15442	(A,B,C,D)
$N(y) * T / P$	SP 4	0.35180	(A,B,C,D)
$M(y) / P$	SP 4	0.21587	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / MC$	SM 4	0.11283	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / MC$	SM 4	1.01070	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / ML$	SM 4	0.11283	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / ML$	SM 4	1.01070	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Rad. Bend. P		3.3	-3.3	3.3	-3.3	3.3	-3.3	3.3	-3.3
Rad. Memb. MC		0.0	0.0	0.0	0.0	-6.0	-6.0	6.0	6.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-19.2	19.2	19.2	-19.2
Rad. Memb. ML		-8.0	-8.0	8.0	8.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-25.6	25.6	25.6	-25.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-29.1	15.4	38.1	-19.8	-20.7	11.0	29.7	-15.4
Tang. Memb. P		5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Tang. Bend. P		18.8	-18.8	18.8	-18.8	18.8	-18.8	18.8	-18.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-125.7	125.7	125.7	-125.7
Tang. Memb. ML		-3.1	-3.1	3.1	3.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-167.6	167.6	167.6	-167.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-146.8	150.8	194.7	-178.2	-104.1	109.7	152.0	-137.1
Shear VC		4.4	4.4	-4.4	-4.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.4	-4.4	4.4	4.4
Shear MT		19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5
Tot. Shear		23.8	23.8	15.1	15.1	15.1	15.1	23.8	23.8
Str. Int.		151.5	154.9	196.1	179.6	106.8	111.9	156.5	141.6

Unitless Prm: $U = 0.45$ TAU = 0.00 (16.32) RHO = 0.00 (2.63)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.14602	(A,B,C,D)
$M(x) / P$	SR 2	0.09324	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / MC$	SR 3	0.14964	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / MC$	SR 3	0.28344	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / ML$	SR 3	0.14964	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / ML$	SR 3	0.28344	(A,B,C,D)
$N(y) * T / P$	SR 2	0.04382	(A,B,C,D)
$M(y) / P$	SR 2	0.02820	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / MC$	SR 3	0.04544	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / MC$	SR 3	0.08388	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / ML$	SR 3	0.04544	(A,B,C,D)

$M(y) * \sqrt{R_m * T} / ML \quad SR \ 3 \quad 0.08388 \quad (A,B,C,D)$

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		13.5	13.5	13.5	13.5	13.5	13.5	13.5	13.5
Rad. Bend. P		51.7	-51.7	51.7	-51.7	51.7	-51.7	51.7	-51.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-31.5	-31.5	31.5	31.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-357.4	357.4	357.4	-357.4
Rad. Memb. ML		-41.9	-41.9	41.9	41.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-476.6	476.6	476.6	-476.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-453.3	396.4	583.8	-472.9	-323.6	287.8	454.1	-364.2
Tang. Memb. P		4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1
Tang. Bend. P		15.6	-15.6	15.6	-15.6	15.6	-15.6	15.6	-15.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-9.5	-9.5	9.5	9.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-105.8	105.8	105.8	-105.8
Tang. Memb. ML		-12.7	-12.7	12.7	12.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-141.0	141.0	141.0	-141.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-134.1	116.7	173.5	-139.9	-95.6	84.6	135.0	-107.8
Shear VC		6.9	6.9	-6.9	-6.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-6.9	-6.9	6.9	6.9
Shear MT		19.3	19.3	19.3	19.3	19.3	19.3	19.3	19.3
Tot. Shear		26.2	26.2	12.4	12.4	12.4	12.4	26.2	26.2
Str. Int.		455.4	398.9	584.1	473.4	324.3	288.5	456.3	366.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		483.5	483.5	483.5	483.5	483.5	483.5	483.5	483.5
Rad. Pl (SUS)		-6.9	-6.9	9.2	9.2	-4.8	-4.8	7.2	7.2
Rad. Q (SUS)		-22.3	22.3	28.9	-28.9	-15.9	15.9	22.5	-22.5
Long. Pm (SUS)		483.5	483.5	483.5	483.5	483.5	483.5	483.5	483.5
Long. Pl (SUS)		2.0	2.0	8.2	8.2	2.8	2.8	7.4	7.4
Long. Q (SUS)		-148.8	148.8	186.4	-186.4	-106.9	106.9	144.5	-144.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.4	4.4	-4.4	-4.4	-4.4	-4.4	4.4	4.4
Shear Q (SUS)		19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5
Pm (SUS)		483.5	483.5	483.5	483.5	483.5	483.5	483.5	483.5
Pm+Pl (SUS)		487.3	487.3	496.6	496.6	488.2	488.2	495.2	495.2
Pm+Pl+Q (Total)		459.0	638.4	679.6	465.2	465.4	595.4	639.9	472.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	483.49	1407.13	Passed



Pm+Pl (SUS)	496.59	2110.70	Passed
Pm+Pl+Q (TOTAL)	679.60	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1
Rad.	Pl (SUS)	-28.4	-28.4	55.4	55.4	-17.9	-17.9	45.0	45.0
Rad.	Q (SUS)	-424.8	424.8	528.3	-528.3	-305.7	305.7	409.2	-409.2
Long.	Pm (SUS)	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1
Long.	Pl (SUS)	-8.7	-8.7	16.8	16.8	-5.5	-5.5	13.6	13.6
Long.	Q (SUS)	-125.4	125.4	156.7	-156.7	-90.1	90.1	121.4	-121.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	6.9	6.9	-6.9	-6.9	-6.9	-6.9	6.9	6.9
Shear	Q (SUS)	19.3	19.3	19.3	19.3	19.3	19.3	19.3	19.3
Pm (SUS)		1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1	1220.1
Pm+Pl (SUS)		1213.6	1213.6	1276.8	1276.8	1217.7	1217.7	1266.5	1266.5
Pm+Pl+Q (Total)		1088.2	1619.0	1804.3	1080.7	1125.2	1508.6	1676.4	1115.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1220.13	1407.13	Passed
Pm+Pl (SUS)	1276.76	2110.70	Passed
Pm+Pl+Q (TOTAL)	1804.27	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N4 100

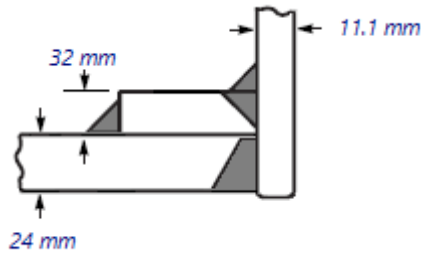
From: 120

Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Elliptical Head	D	4650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1202.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1202.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N4 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.24 \cdot 4656.0 \cdot 0.998 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
 $= 20.2267 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 49.02 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 0.5020 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	256.4701 mm
Parallel to Vessel Wall, opening length	d	128.2350 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1202.2 / 1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1202.2 / 1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1406.1 / 1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.854, 0.999)$
 $= 0.854$

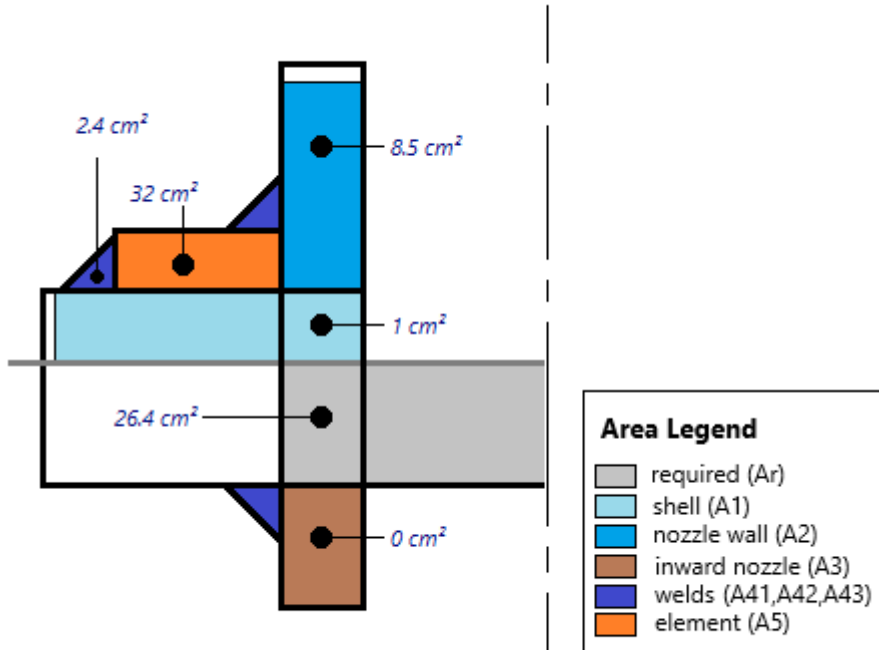


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.416	NA	NA
Area in Shell	A1	0.973	NA	NA
Area in Nozzle Wall	A2	8.500	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.407	NA	NA
Area in Element	A5	31.977	NA	NA
TOTAL AREA AVAILABLE	Atot	43.858	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 49.87 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	159.7573	32.0000 mm
Based on given Pad Diameter:	214.3000	14.5463 mm
Based on the Estimated Diameter Limit:	254.8826	13.8018 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (128.235 \cdot 20.2267 \cdot 1.0 + 2 \cdot 8.1252 \cdot 20.2267 \cdot 1.0 \cdot (1 - 0.85))$
 $= 26.416 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 128.235(1.0 \cdot 21.0 - 1.0 \cdot 20.227) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 20.2267) \cdot (1 - 0.854)$
 $= 0.973 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.5) \cdot 0.8544 / \sin(53.3)$



= 8.500 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (1.) • 0.85 + (0.0) • 0.85 + 155.3766² • 1.
 = 2.407 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (249.4883 - 149.4883) • 32.0 • 0.9993
 = 31.977 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24.0, tr = 20.227, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.963, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.24/19.99 = 0.612



Weld Size Calculations, Description: N4 100

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6876 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	16.9680 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (26.4164 - 0.9733 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 20.2267)) \cdot 1407) \\ &= 35952.90 \text{ kgf} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (8.5 + 31.9774 + 2.407 - 0.0 \cdot 0.85) \cdot 1407 \\ &= 60344.09 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (8.5 + 0.0 + 0.8544 + (2.9156)) \cdot 1407 \\ &= 17265.47 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (8.5 + 0.0 + 2.407 + 31.9774 + (2.9156)) \cdot 1407 \\ &= 64446.70 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 149.4883 \cdot 10.0 \cdot 0.49 \cdot 1202 \\ &= 13832. \text{ kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 214.3 \cdot 24.0 \cdot 0.49 \cdot 1406 \\ &= 55664. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 69.4308) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202 \\ &= 14915. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 149.4883 \cdot 32.0 \cdot 0.74 \cdot 1406 \\ &= 78187. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 149.4883 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407 \\ &= 51347. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (55664 + 14915) = 70579 \text{ kgf}$$

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (13832 + 78187 + 51347 + 0) = 143367 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (55664 + 51347 + 0) = 107011 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 70578 kgf, must exceed W = 35952 kgf or W1 = 60344 kgf
Path 2-2 = 143366 kgf, must exceed W = 35952 kgf or W2 = 17265 kgf
Path 3-3 = 107011 kgf, must exceed W = 35952 kgf or W3 = 64446 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N4 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $12.24 \cdot 4656.0 \cdot 0.998 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
= 20.2267 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.24 \cdot 49.02 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
= 0.5020 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992	mm
Parallel to Vessel Wall, opening length	d	98.0496	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

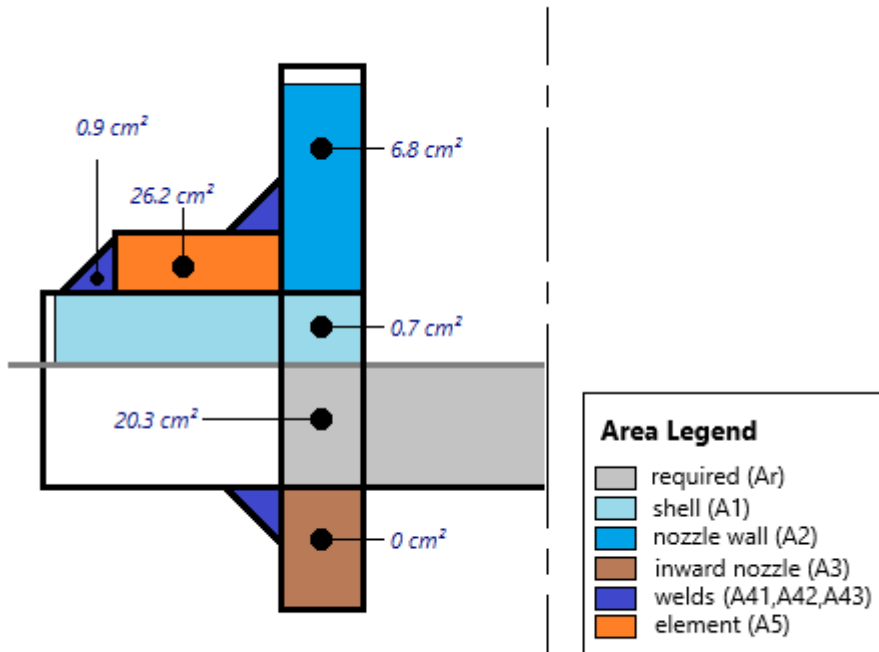


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	20.311	NA	NA
Area in Shell	A1	0.740	NA	NA
Area in Nozzle Wall	A2	6.814	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.854	NA	NA
Area in Element	A5	26.157	NA	NA
TOTAL AREA AVAILABLE	Atot	34.566	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	151.5214	32.0000 mm
Based on given Pad Diameter:	214.3000	14.5611 mm
Based on the Estimated Diameter Limit:	194.5117	14.8493 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (98.0496 \cdot 20.2267 \cdot 1.0 + 2 \cdot 8.1252 \cdot 20.2267 \cdot 1.0 \cdot (1 - 0.85))$
 $= 20.311 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 98.05(1.0 \cdot 21.0 - 1.0 \cdot 20.227) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 20.2267) \cdot (1 - 0.854)$
 $= 0.740 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.5) \cdot 0.8544$



$$= 6.814 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.85 + (0.0)^2 \cdot 0.85 + 0.0^2 \cdot 1.$
 $= 0.854 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (196.0992 - 114.3) \cdot 32.0 \cdot 0.9993$
 $= 26.157 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5020 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 23.2267 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 23.2267 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(23.2267, 4.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.502, 8.2578)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Weld Size Calculations, Description: N4 100

Intermediate Calc. for nozzle/shell Welds	T_{min}	8.1252 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (20.3109 - 0.7399 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 20.2267)) / 1407)$
 $= 27690.07 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr2) \cdot S_v$
 $= (6.8142 + 26.1572 + 0.8544 - 0.0 \cdot 0.85) \cdot 1407$
 $= 47597.43 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$
 $= (6.8142 + 0.0 + 0.8544 + (2.9156)) \cdot 1407$
 $= 14893.34 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (6.8142 + 0.0 + 0.8544 + 26.1572 + (2.9156)) \cdot 1407$
 $= 51700.05 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 1202 \\ &= 10576. \text{ kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 214.3 \cdot 24.0 \cdot 0.49 \cdot 1406 \\ &= 55664. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 53.0874) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202 \\ &= 11404. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 114.3 \cdot 32.0 \cdot 0.74 \cdot 1406 \\ &= 59783. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 114.3 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407 \\ &= 39260. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (55664 + 11404) = 67068 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10576 + 59783 + 39260 + 0) = 109619 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (55664 + 39260 + 0) = 94925 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 67068	kgf, must exceed W = 27690 kgf	or W1 = 47597 kgf
Path 2-2 = 109619	kgf, must exceed W = 27690 kgf	or W2 = 14893 kgf
Path 3-3 = 94924	kgf, must exceed W = 27690 kgf	or W3 = 51700 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 31.4841 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 558.7468 mm

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Input, Nozzle Desc: N5 100

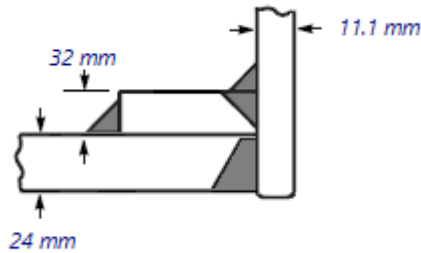
From: 120

Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Elliptical Head	D	4650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1202.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1202.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N5 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.24 \cdot 4656.0 \cdot 0.998 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
 $= 20.2267 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 49.02 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 0.5020 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	256.4701 mm
Parallel to Vessel Wall, opening length	d	128.2350 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1202.2 / 1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1202.2 / 1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1406.1 / 1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.854, 0.999)$
 $= 0.854$

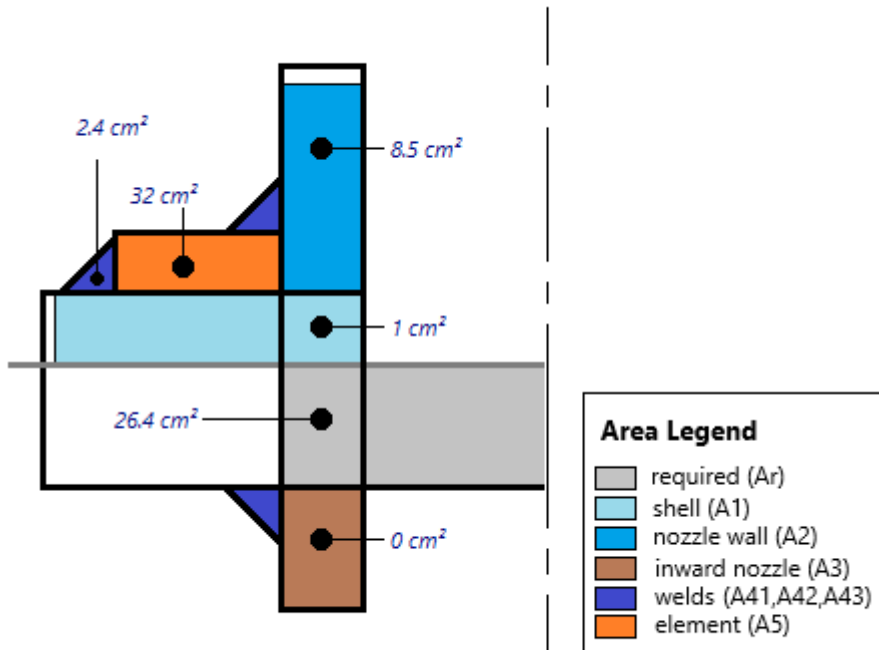


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.416	NA	NA
Area in Shell	A1	0.973	NA	NA
Area in Nozzle Wall	A2	8.500	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.407	NA	NA
Area in Element	A5	31.977	NA	NA
TOTAL AREA AVAILABLE	Atot	43.858	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 49.87 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	159.7573	32.0000 mm
Based on given Pad Diameter:	214.3000	14.5463 mm
Based on the Estimated Diameter Limit:	254.8826	13.8018 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (128.235 \cdot 20.2267 \cdot 1.0 + 2 \cdot 8.1252 \cdot 20.2267 \cdot 1.0 \cdot (1 - 0.85))$
 $= 26.416 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 128.235(1.0 \cdot 21.0 - 1.0 \cdot 20.227) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 20.2267) \cdot (1 - 0.854)$
 $= 0.973 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.5) \cdot 0.8544 / \sin(53.3)$



= 8.500 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.85 + (0.0) • 0.85 + 155.3766² • 1.
= 2.407 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (249.4883 - 149.4883) • 32.0 • 0.9993
= 31.977 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24.0, tr = 20.227, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.963, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.502, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.075, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 12.24/19.99 = 0.612



Weld Size Calculations, Description: N5 100

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6876 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	16.9680 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (26.4164 - 0.9733 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 20.2267)) \cdot 1407)$$

$$= 35952.90 \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (8.5 + 31.9774 + 2.407 - 0.0 \cdot 0.85) \cdot 1407$$

$$= 60344.09 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (8.5 + 0.0 + 0.8544 + (2.9156)) \cdot 1407$$

$$= 17265.47 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (8.5 + 0.0 + 2.407 + 31.9774 + (2.9156)) \cdot 1407$$

$$= 64446.70 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 149.4883 \cdot 10.0 \cdot 0.49 \cdot 1202$$

$$= 13832. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 214.3 \cdot 24.0 \cdot 0.49 \cdot 1406$$

$$= 55664. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 69.4308) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202$$

$$= 14915. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 149.4883 \cdot 32.0 \cdot 0.74 \cdot 1406$$

$$= 78187. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 149.4883 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407$$

$$= 51347. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (55664 + 14915) = 70579 \text{ kgf}$$

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (13832 + 78187 + 51347 + 0) = 143367 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (55664 + 51347 + 0) = 107011 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 70578 kgf, must exceed W = 35952 kgf or W1 = 60344 kgf
Path 2-2 = 143366 kgf, must exceed W = 35952 kgf or W2 = 17265 kgf
Path 3-3 = 107011 kgf, must exceed W = 35952 kgf or W3 = 64446 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N5 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $12.24 \cdot 4656.0 \cdot 0.998 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
= 20.2267 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.24 \cdot 49.02 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
= 0.5020 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm
Parallel to Vessel Wall, opening length	d	98.0496 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

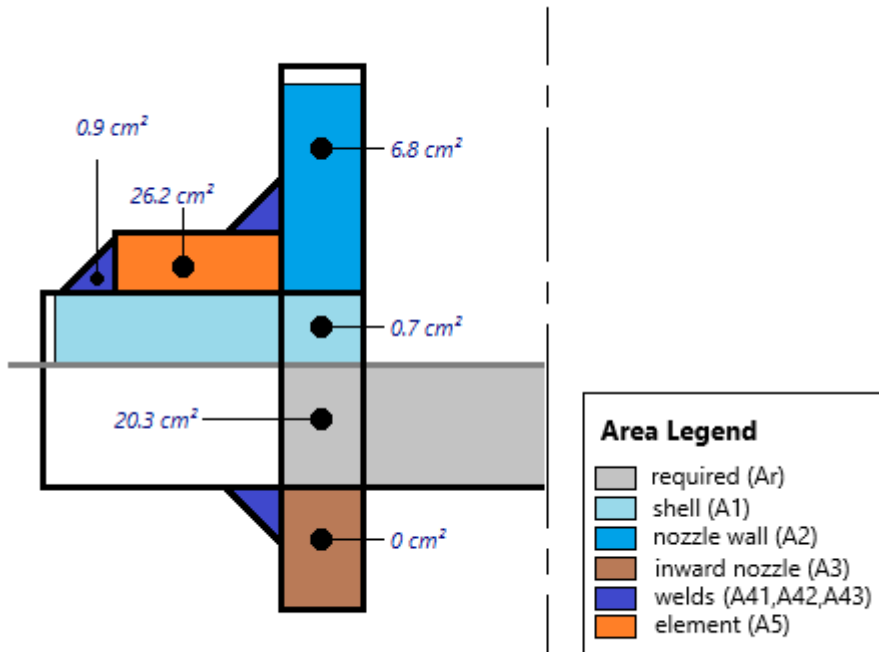


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	20.311	NA	NA
Area in Shell	A1	0.740	NA	NA
Area in Nozzle Wall	A2	6.814	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.854	NA	NA
Area in Element	A5	26.157	NA	NA
TOTAL AREA AVAILABLE	Atot	34.566	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 151.5214 32.0000 mm
 Based on given Pad Diameter: 214.3000 14.5611 mm
 Based on the Estimated Diameter Limit: 194.5117 14.8493 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (98.0496 \cdot 20.2267 \cdot 1.0 + 2 \cdot 8.1252 \cdot 20.2267 \cdot 1.0 \cdot (1 - 0.85))$
 $= 20.311 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 98.05(1.0 \cdot 21.0 - 1.0 \cdot 20.227) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 20.2267) \cdot (1 - 0.854)$
 $= 0.740 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.5) \cdot 0.8544$



$$= 6.814 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.85 + (0.0)^2 \cdot 0.85 + 0.0^2 \cdot 1.$
 $= 0.854 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (196.0992 - 114.3) \cdot 32.0 \cdot 0.9993$
 $= 26.157 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5020 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 23.2267 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 23.2267 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(23.2267, 4.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.502, 8.2578)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Weld Size Calculations, Description: N5 100

Intermediate Calc. for nozzle/shell Welds	T_{min}	8.1252 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (20.3109 - 0.7399 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 20.2267)) / 1407)$
 $= 27690.07 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr2) \cdot Sv$
 $= (6.8142 + 26.1572 + 0.8544 - 0.0 \cdot 0.85) \cdot 1407$
 $= 47597.43 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (6.8142 + 0.0 + 0.8544 + (2.9156)) \cdot 1407$
 $= 14893.34 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (6.8142 + 0.0 + 0.8544 + 26.1572 + (2.9156)) \cdot 1407$
 $= 51700.05 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 1202 \\ &= 10576. \text{ kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 214.3 \cdot 24.0 \cdot 0.49 \cdot 1406 \\ &= 55664. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 53.0874) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202 \\ &= 11404. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 114.3 \cdot 32.0 \cdot 0.74 \cdot 1406 \\ &= 59783. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 114.3 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407 \\ &= 39260. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (55664 + 11404) = 67068 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10576 + 59783 + 39260 + 0) = 109619 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (55664 + 39260 + 0) = 94925 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 67068	kgf, must exceed W = 27690 kgf	or W1 = 47597 kgf
Path 2-2 = 109619	kgf, must exceed W = 27690 kgf	or W2 = 14893 kgf
Path 3-3 = 94924	kgf, must exceed W = 27690 kgf	or W3 = 51700 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 31.4841 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 558.7468 mm

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Input, Nozzle Desc: N6 80

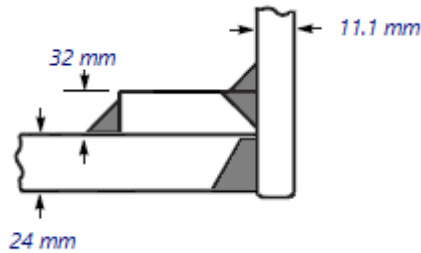
From: 120

Pressure for Reinforcement Calculations	P	12.236	kgf/cm ²
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Elliptical Head	D	4650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1202.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1202.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1406.14	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1406.14	kgf/cm ²
Diameter of Pad along vessel surface	Dp	168.9000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		40.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N6 80

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.24 \cdot 4656.0 \cdot 0.8989 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
 $= 18.2129 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.24 \cdot 36.32 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
 $= 0.3720 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.6400 mm
Parallel to Vessel Wall, opening length	d	78.8200 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1202.2/1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1202.2/1407.1)$
 $= 0.854$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 1406.1/1407.1)$
 $= 0.999$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.854, 0.999)$
 $= 0.854$

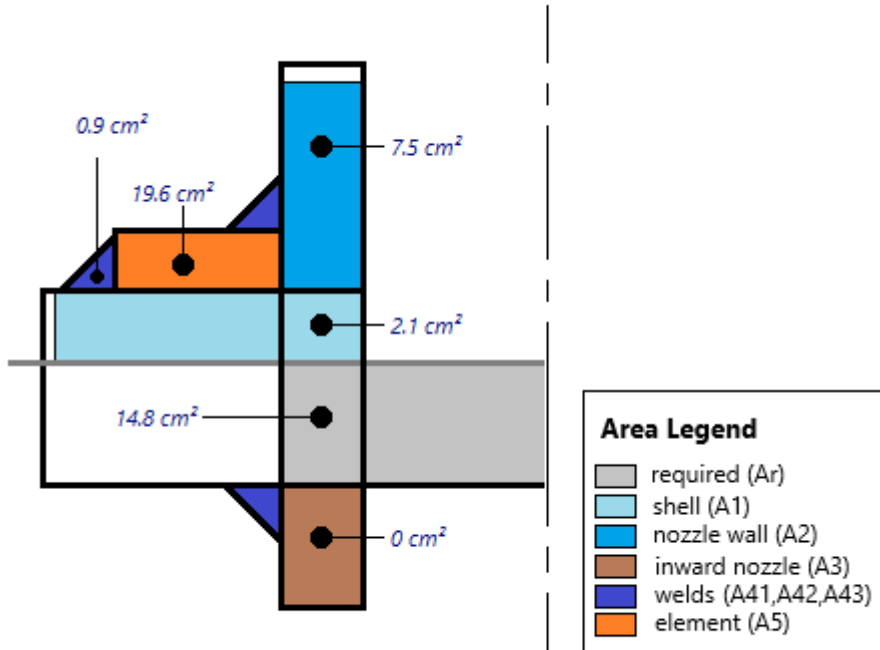


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.786	NA	NA
Area in Shell	A1	2.131	NA	NA
Area in Nozzle Wall	A2	7.460	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.854	NA	NA
Area in Element	A5	19.567	NA	NA
TOTAL AREA AVAILABLE	Atot	30.012	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 67.18 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 102.4757 32.0000 mm
 Based on given Pad Diameter: 168.9000 7.0996 mm
 Based on Shell or Nozzle Thickness: 135.4992 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.82 \cdot 18.2129 \cdot 1.0 + 2 \cdot 8.1252 \cdot 18.2129 \cdot 1.0 \cdot (1 - 0.85))$
 $= 14.786 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.82(1.0 \cdot 21.0 - 1.0 \cdot 18.213) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 18.2129) \cdot (1 - 0.854)$
 $= 2.131 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.37) \cdot 0.8544 / \sin(68.3)$



= 7.460 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 0.85 + (0.0)^2 \cdot 0.85 + 0.0^2 \cdot 1.$
= 0.854 cm²

Area Available in Element [A5]:
= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$
= $(157.64 - 96.4506) \cdot 32.0 \cdot 0.9993$
= 19.567 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.735, tr = 0.372, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.735, tr = 0.372, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.735, tr = 0.372, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24.0, tr = 18.213, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.867$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-9 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.372, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.24/19.99 = 0.612$



Weld Size Calculations, Description: N6 80

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6876 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	16.9680 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
= $\max(0, (14.7864 - 2.1308 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 18.2129)) / 1407)$
= 18352.61 kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
= $(7.4601 + 19.5668 + 0.8544 - 0.0 \cdot 0.85) \cdot 1407$
= 39232.56 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
= $(7.4601 + 0.0 + 0.8544 + (2.9156)) \cdot 1407$
= 15802.14 kgf

Weld Load [W3]:
= $(A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
= $(7.4601 + 0.0 + 0.8544 + 19.5668 + (2.9156)) \cdot 1407$
= 43335.18 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
= $(3.1416/2.0) \cdot 96.4506 \cdot 10.0 \cdot 0.49 \cdot 1202$
= 8925. kgf

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 168.9 \cdot 24.0 \cdot 0.49 \cdot 1406$
= 43872. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
= $(3.1416 \cdot 43.8177) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202$
= 9413. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 96.4506 \cdot 32.0 \cdot 0.74 \cdot 1406$
= 50447. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
= $(3.1416/2.0) \cdot 96.4506 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407$
= 33129. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (43872 + 9413) = 53284$ kgf

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (8925 + 50447 + 33129 + 0) = 92501 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (43872 + 33129 + 0) = 77001 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 53284 kgf, must exceed W = 18352 kgf or W1 = 39232 kgf
Path 2-2 = 92500 kgf, must exceed W = 18352 kgf or W2 = 15802 kgf
Path 3-3 = 77000 kgf, must exceed W = 18352 kgf or W3 = 43335 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N6 80

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.624 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $12.24 \cdot 4656.0 \cdot 0.8989 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 12.24)$
= 18.2129 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.24 \cdot 36.32 / (1202 \cdot 1.0 - 0.6 \cdot 12.24)$
= 0.3720 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2992	mm
Parallel to Vessel Wall, opening length	d	72.6496	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.3130	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

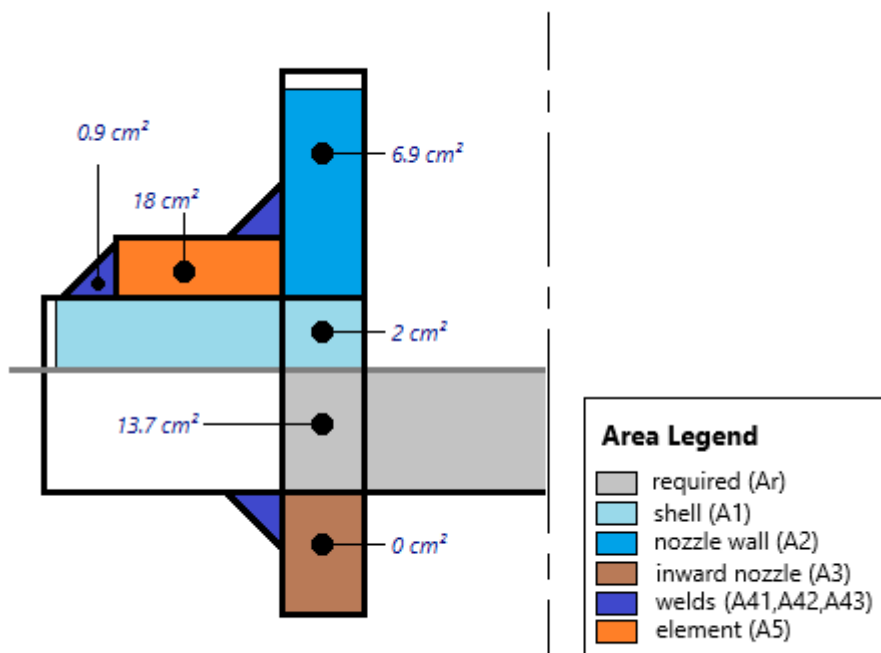


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.663	NA	NA
Area in Shell	A1	1.959	NA	NA
Area in Nozzle Wall	A2	6.930	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.854	NA	NA
Area in Element	A5	18.035	NA	NA
TOTAL AREA AVAILABLE	Atot	27.779	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 101.1553 32.0000 mm
 Based on given Pad Diameter: 168.9000 6.9535 mm
 Based on Shell or Nozzle Thickness: 124.1506 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.6496 \cdot 18.2129 \cdot 1.0 + 2 \cdot 8.1252 \cdot 18.2129 \cdot 1.0 \cdot (1 - 0.85))$
 $= 13.663 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.65(1.0 \cdot 21.0 - 1.0 \cdot 18.213) - 2 \cdot 8.125$
 $(1.0 \cdot 21.0 - 1.0 \cdot 18.2129) \cdot (1 - 0.854)$
 $= 1.959 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 52.31) \cdot (8.13 - 0.37) \cdot 0.8544$



$$= 6.930 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.85 + (0.0)^2 \cdot 0.85 + 0.0^2 \cdot 1.$
 $= 0.854 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (145.2992 - 88.9) \cdot 32.0 \cdot 0.9993$
 $= 18.035 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.3720 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 23.2267 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 23.2267 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 7.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(23.2267, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.372, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Weld Size Calculations, Description: N6 80

Intermediate Calc. for nozzle/shell Welds	T_{min}	8.1252 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (13.6626 - 1.9589 + 2 \cdot 8.1252 \cdot 0.8544 \cdot (1.0 \cdot 21.0 - 18.2129)) / 1407)$
 $= 17013.26 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr2) \cdot Sv$
 $= (6.9305 + 18.035 + 0.8544 - 0.0 \cdot 0.85) \cdot 1407$
 $= 36331.93 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (6.9305 + 0.0 + 0.8544 + (2.9156)) \cdot 1407$
 $= 15056.93 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (6.9305 + 0.0 + 0.8544 + 18.035 + (2.9156)) \cdot 1407$
 $= 40434.54 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 1202 \\ &= 8226. \text{ kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 168.9 \cdot 24.0 \cdot 0.49 \cdot 1406 \\ &= 43872. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 40.3874) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 1202 \\ &= 8676. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 88.9 \cdot 32.0 \cdot 0.74 \cdot 1406 \\ &= 46498. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 88.9 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 1407 \\ &= 30536. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (43872 + 8676) = 52547 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (8226 + 46498 + 30536 + 0) = 85259 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (43872 + 30536 + 0) = 74407 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 52547	kgf, must exceed W = 17013 kgf	or W1 = 36331 kgf
Path 2-2 = 85259	kgf, must exceed W = 17013 kgf	or W2 = 15056 kgf
Path 3-3 = 74407	kgf, must exceed W = 17013 kgf	or W3 = 40434 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 17.2660 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 542.9462 mm

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